

# **Talking Through Trouble: The Strategic Attention of CEO Letters in the Face of ESG Decline**



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*Master's Thesis*

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## **Abstract**

CEO shareholder letters are discretionary, unaudited texts in which executives choose not only how to write but what to write about. This study asks whether a decline in ESG performance leads CEOs to shift their communicative attention toward ESG, and whether ownership structure conditions this response. It draws on the attention-based view, which holds that executive attention is finite and must be selectively allocated, together with legitimacy theory. The hypotheses were tested on a panel of 150 large US-listed firms observed from 2018 to 2022, giving an estimation sample of 596 firm-years. No statistically significant relationship is found between an ESG decline and ESG attention, and neither institutional nor firm-oriented ownership moderates the relationship. The null result holds across alternative functional forms, treatments of outliers, subsamples, and an alternative timing of the score change. Although theory gives strong reasons to expect CEOs to devote more attention to ESG after a decline, no such shift is found; this clearly identified null is the study's central contribution, together with a novel method for studying impression management through the distribution of topic attention rather than sentiment.

Keywords: CEO shareholder letters, Impression management, Attention-based view, Legitimacy theory, ESG decline, Ownership structure, Content analysis, Topic attention.

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## 1. Introduction

CEO letters to shareholders are increasingly viewed as deliberate instruments of strategic communication rather than neutral accounts of firm performance (Boudt & Thewissen, 2019). Published annually and signed by the CEO, they are typically included in the annual report and provide shareholders and other stakeholders with information about the company's strategic goals, business performance, and future outlook (Patelli & Pedrini, 2014). Unlike the audited and highly regulated sections of the annual report, CEO letters are neither audited nor subject to a prescribed format. As a result, they are one of the few disclosure channels in which the content largely reflects managerial discretion over what is communicated and how it is communicated (Brennan & Merkl-Davies, 2013). This discretionary character makes CEO letters particularly useful for examining how executives allocate communicative attention in their communication with stakeholders.

A substantial body of research examines how firms and their executives communicate through the tone and sentiment of financial disclosures, including earnings announcements and CEO letters (Davis et al., 2012; Boudt & Thewissen, 2019). Because CEO letters are discretionary and unaudited, the agency conflict between managers and shareholders (Jensen & Meckling, 1976) can become visible in the form of impression management. Impression management refers to the strategic shaping of communication to influence stakeholders' perceptions rather than to disclose information in a neutral manner (Hooghiemstra, 2000; Patelli & Pedrini, 2014). Boudt and Thewissen (2019) provide a concrete example: CEOs deliberately place positive words later in the letter, causing net sentiment to become more positive toward the end and leaving readers with a more favorable final impression than the firm's performance alone would justify.

This literature primarily examines tone, sentiment, and framing. It therefore focuses on how CEOs communicate, while paying much less attention to what they choose to communicate (Bassyouny et al., 2022). That distinction matters because the substance of communication can be managed just as the style of communication can. A CEO can shape impressions not only through the way a topic is presented, but also through the decision to emphasize some topics while giving little or no attention to others. In the impression-management literature, this type of choice is described as thematic manipulation (Brennan & Merkl-Davies, 2013). The concept is extended here from an emphasis on favorable over unfavorable topics to the allocation of attention across topics.

Topic selection is a consequential strategic choice for two reasons. First, the CEO letter is a bounded space, so giving more attention to one topic necessarily leaves less room for another. Emphasizing one subject is therefore also a decision to downplay another. Second, attention itself is limited. The attention-based view holds that organizational attention is scarce and exercised through decision-makers (Ocasio, 1997), meaning that what a CEO foregrounds reflects what is being prioritized. Cong, Freedman, and Park (2014) provide empirical support for this view, showing that CEO letters devote more attention to environmental matters when environmental performance is weaker. This is consistent with the broader finding that weaker performers disclose more, rather than less, in an effort to manage their standing (Patten, 2002).

If topic emphasis responds to circumstances, a decline in ESG performance provides a useful setting in which to study it. Such a decline confronts the CEO with a choice: address the weakness directly or direct attention toward other issues. This study focuses on that trigger and operationalizes it as a decline in the firm's ESG rating. ESG is an appropriate trigger for three reasons. First, the rating is produced by an external agency and is publicly visible, so a decline represents an authoritative signal that stakeholders can observe independently of the CEO (Gillan et al., 2021). Second, a decline threatens the firm's legitimacy by creating a gap between how the firm is perceived and how stakeholders expect it to behave, which increases pressure to respond (Suchman, 1995). Third, this pressure leaves a measurable trace because attention is finite. More space devoted to ESG can therefore be interpreted as a reallocation of attention away from other topics.

The study examines four topics that compete for the CEO's limited communicative space, with each representing a distinct claim on that space. ESG is the domain in which the trigger occurs. Financial performance is the topic that typically dominates CEO letters (Conaway & Wardrope, 2010). Innovation offers a forward-looking alternative to emphasizing present weaknesses (Merkley, 2014), while diversity represents an adjacent and independently salient domain in the US context (Hellerstedt, Uman & Wennberg, 2024). The theoretical framework developed in the next section explains how CEOs may reallocate attention across these competing topics when ESG performance deteriorates.

Existing research has not brought these strands together. Studies of CEO letters have documented patterns in topic emphasis, but they have generally not examined what causes the relative weight of those topics to change over time (Conaway & Wardrope, 2010). When topic-level change has been analyzed, researchers have usually focused on a single topic, such as

environmental rhetoric, rather than examining how attention is redistributed across several competing topics. ESG has become a prominent and separately scrutinized dimension of firm performance (Gillan et al., 2021), yet no study has isolated an ESG decline as a trigger of communicative reallocation or examined whether ownership structure influences how CEOs respond. These gaps are important. If CEOs systematically reallocate attention across topics when ESG performance deteriorates, sentiment-based measures alone cannot explain how shareholder letters are constructed. In that case, stakeholders may be influenced not only by what is said, but also by which topics receive attention and which are largely omitted.

Ownership is a fundamental layer of corporate governance: owners appoint the board and shape the incentives within which CEOs operate (Shleifer & Vishny, 1997). This is why the study focuses on ownership rather than on board composition or CEO characteristics. Owners are not homogeneous; they differ in their objectives, monitoring capacity, and access to firm information (Connelly et al., 2010; Collin et al., 2017). These differences may shape the extent to which the CEO letter serves as a channel for legitimacy repair (Suchman, 1995). Institutional investors pursue financially oriented objectives, relying on public disclosure and standardized ratings rather than private access to management. The CEO letter should therefore matter more to them, making a stronger reallocation toward ESG more likely. Firm-oriented owners, including founding families, individuals, and non-financial corporations with long-term holdings, have more direct access to management and evaluate the firm based on long-term objectives rather than ratings. For them, the CEO letter should matter less, and the reallocation should therefore be weaker. Whether ownership conditions this relationship remains an open empirical question that this study examines (Collin et al., 2017; Federo et al., 2020).

Together, these gaps point to a specific and unaddressed research need. This study therefore addresses the following research question:

*How does a decline in ESG performance reshape the allocation of communicative attention across topics in CEO shareholder letters, and how does ownership structure moderate this relationship?*

The study addresses this question using a panel of US-listed firms covering the period from 2018 to 2022. CEO shareholder letters are collected manually from corporate annual reports and analyzed through content analysis. Communicative attention is measured as the share of letter text devoted to each of the four topics. ESG performance change is operationalized as the



year-on-year percentage change in a firm's ESG score, while ownership structure is captured through institutional ownership and firm-oriented ownership.

The main findings are straightforward. No statistically significant relationship is found between an ESG decline and the attention CEOs devote to ESG, either in absolute or relative terms. Institutional ownership and firm-oriented ownership also do not moderate this relationship. The null result remains consistent across alternative functional forms, different treatments of outliers, subsample analyses, and an alternative timing of the ESG score change. One pattern differs from the theoretical expectation: firms experiencing the most severe ESG declines are the least likely to mention ESG at all. This result is exploratory because it does not remain robust once the analysis is aligned with the timing at which CEOs could have observed the decline, and its direction cannot be established with confidence. These findings are examined through a topic-attention approach that treats impression management as a matter of which topics receive attention rather than how positively they are phrased. Read through this lens, a well-identified null result against a strong theoretical expectation is informative in its own right.

This study makes two contributions. First, it reframes impression management as a question of attention allocation. This is a theoretical choice about what to measure rather than a technique for measuring it. Instead of focusing on the tone of a letter or the emphasis given to a single domain in isolation, the study examines how a CEO distributes finite attention across several competing strategic topics: ESG, financial performance, innovation, and diversity. The CEO letter is a bounded space in which emphasizing one topic necessarily leaves less room for the others. Prior work has largely examined tone and sentiment (Boudt & Thewissen, 2019) or how firms communicate rather than which topics they foreground (Bassyouny et al., 2022). The relational, allocation-based lens developed here therefore provides a distinct starting point for studying CEO communication and remains useful whether or not a given study finds an effect.

Second, viewed through this lens, the study reports a null result against a strong theoretical expectation. The contribution lies less in the research question itself than in how this null result is interpreted. The prediction was based on two convergent mechanisms, automatic vigilance and deliberate legitimacy repair, as well as on prior evidence that firms communicate more about a domain when their performance in that domain is weak (Patten, 2002; Cong et al., 2014). A null result is more informative when it contradicts a well-supported expectation than when it fails to support a weak one, and null results for hypotheses with strong prior odds are

less likely to be false alarms than those for implausible hypotheses (Roloff & Zyphur, 2019). At the same time, the analysis does not justify the stronger conclusion that CEOs demonstrably do not respond. A non-significant result may reflect limited statistical power rather than the true absence of an effect (Cashen & Geiger, 2004), and because attention is captured through keyword-based topic shares, a genuine but subtle reallocation cannot be ruled out. The contribution is therefore primarily interpretive. The reallocation toward ESG predicted by theory does not appear in the data, and the study treats this absence as a meaningful finding rather than a failed test. This interpretation is further supported by the ownership results: even institutional owners, who are expected to be most attentive to standardized ESG ratings, do not strengthen the relationship. The study discusses several plausible explanations, including the gradual and diffuse nature of ESG score changes relative to discrete events, the partly templated character of CEO letter writing, and a response horizon that may extend beyond the immediately following letter.

The remainder of this thesis is organized as follows. Chapter 2 develops the theoretical framework and presents the hypotheses. Chapter 3 describes the sample, the data, the measurement of the variables, and the estimation strategy. Chapter 4 presents the empirical results. Chapter 5 reports robustness checks. Chapter 6 discusses the findings and concludes with theoretical and practical implications, limitations, and directions for future research.

## **2. Theoretical Background and Hypotheses Development**

### **2.1 The Attention-Based View**

How CEOs allocate attention across the issues competing for their notice is explained by the Attention-Based View (ABV), which therefore provides the central theoretical foundation of this study. Originating with Ocasio's (1997) foundational work, the ABV argues that organizational behavior depends on how firms channel and allocate the limited attention of their decision-makers. Executives have limited cognitive capacity and cannot attend to all issues simultaneously. As a result, their attention is selective and context dependent. It is directed toward the issues that stand out in the situation the firm currently faces and shaped by the firm's rules, resources, and relationships. Issues that are salient, that affect the firm's standing, or that are brought to executives' attention by influential stakeholders are much more

likely to receive attention than issues that remain in the background. Managerial action therefore depends on where attention is directed. Because attention is scarce, its allocation is inherently strategic: giving more attention to one issue necessarily leaves less attention for others. The emphasis executives place on particular issues thus reveals and helps shape the firm's priorities. This matters beyond the individual executive, since top managers exert substantial influence over firm-level outcomes (Finkelstein, Hambrick & Cannella, 2009).

If executive priorities are reflected in what CEOs choose to communicate, then the allocation of executive attention should also be observable in corporate communication. The following section develops the CEO letter as such a window.

## **2.2 The CEO Letter as a Window into Executive Attention**

Among the many texts firms produce, the CEO shareholder letter occupies a distinctive position for studying executive attention. It is published annually, signed by the CEO, and addressed directly to the firm's stakeholders, making it one of the most visible and personal forms of corporate communication (Patelli & Pedrini, 2014). Unlike the audited sections of the annual report, CEO letters are neither regulated in content nor prescribed in form. Their content therefore reflects deliberate choices about what to emphasize and what to leave aside (Brennan & Merkl-Davies, 2013). This discretionary character makes the letter particularly informative for analysis. Because the CEO letter is limited in length, giving more attention to one topic necessarily leaves less room for another. The overall composition of the letter therefore reflects the CEO's priorities at the time it is written.

This characteristic aligns closely with the central claim of the Attention-Based View that attention is limited and selectively directed. If executives emphasize certain issues because those issues command their attention, then the topics that receive more space in a CEO letter, and those that receive less, should reflect the underlying allocation of executive attention at the time of writing. Ocasio (1997) addresses this apparent tension through the concept of situated attention: decision-makers focus on the issues triggered by the situation they face, regardless of whether that attention operates through controlled or automatic processes. The issues emphasized in a CEO letter are therefore not independent of circumstances but shaped by them, so deliberate emphasis and situational responsiveness describe the same underlying process rather than competing explanations. Prior research supports this interpretation. Studies examining how shareholder letter content changes around major external events show that

letter composition tracks corresponding shifts in managerial priorities, indicating that the letter responds to changing circumstances rather than functioning as a fixed corporate ritual (Cho & Hambrick, 2006; Eggers & Kaplan, 2009). The CEO letter is therefore both discretionary and sensitive to changing circumstances, and this combination makes it a useful window into how executives allocate attention when the environment changes.

The remaining question is which circumstances are consequential enough to shift executive attention. The next section develops an ESG rating decline as one such circumstance.

### **2.3 ESG Rating Declines as an Attentional Trigger**

The previous section argued that the CEO letter reflects shifts in managerial attention. The next question is what causes those shifts. The attention-based view suggests that executives are drawn to issues that are salient, especially when they are negative and urgent (Brielmaier & Friesl, 2023). An ESG rating decline fits this description. Its implications extend beyond salience, however, because it can also threaten the firm's legitimacy and therefore its standing with external stakeholders. To explain why an ESG rating decline creates pressure for CEOs, the idea of organizational legitimacy must first be introduced.

Legitimacy theory argues that organizations depend on being perceived as acting in accordance with the norms and expectations of the society in which they operate (Suchman, 1995). This perception is not merely symbolic. It affects the firm's access to capital, stakeholder relationships, and long-term survival. A decline in a firm's ESG rating signals to external stakeholders that the firm is falling short of prevailing expectations in the environmental, social, and governance domain. The signal is particularly consequential because ESG ratings are produced by external agencies, are publicly visible, and are closely monitored by investors alongside financial performance (Gillan et al., 2021). A decline is therefore not a signal that management can absorb quietly. It is an externally assigned and authoritative assessment that stakeholders can observe before the CEO has an opportunity to respond. The result is a gap between how the firm is perceived and how stakeholders expect it to behave, creating legitimacy pressure that management must address (Suchman, 1995; Bansal & Clelland, 2004).

The CEO letter is particularly relevant in this context. It is discretionary, unaudited, and directed at the owners whose perceptions are most important to management, making it a useful channel through which CEOs can express awareness of and concern about a legitimacy threat

(Arvidsson, 2023). One feature of this channel shapes what a response can look like. Because the letter is written and published before the ESG score for the current fiscal year is finalized, the most recent change a CEO can actually observe when writing is the one completed in the prior year. The argument therefore concerns the CEO's response to the most recently observable movement in the firm's ESG standing. It does not concern a contemporaneous verdict that does not yet exist. This distinction is reflected in the empirical design, which also estimates the models using the lagged, observable change. The following hypotheses develop how this pressure may be reflected in the composition of the letter.

## **2.4 Four Topics as Competing Claims on Attention**

The ABV suggests that emphasis in a CEO letter is inherently comparative because topics compete for a limited amount of attention. The importance of any one topic can therefore be understood only in relation to the others. Since the CEO letter is limited in length, giving more attention to one topic necessarily leaves less room for competing topics. This study focuses on ESG because it is the domain in which the trigger examined here occurs, and it asks how the emphasis placed on ESG changes relative to the topics competing with it when a firm's ESG standing is called into question.

This comparison is important because the length and composition of CEO letters can vary from year to year. More ESG language could indicate a genuine shift in attention away from other topics, but it could also reflect a longer letter overall. Examining ESG in isolation cannot distinguish between these possibilities. Identifying a reallocation of attention therefore requires observing the competing topics directly. Alongside ESG, three additional topics form the relevant comparison: financial performance, which typically dominates CEO letters (Conaway & Wardrope, 2010); diversity, which reflects a closely related but distinct legitimacy concern; and innovation, the most plausible alternative destination for attention when a firm seeks to redirect it. These topics do not capture the full content of a CEO letter, which also includes greetings, macroeconomic commentary, leadership communication, and organizational announcements. They do, however, represent the main arena in which a response to an ESG legitimacy threat would be expected to become visible.

Financial performance is the topic that typically dominates this space. Across organizational and cultural contexts, it consistently receives the largest share of CEO letter content (Conaway & Wardrope, 2010), and the letter is conventionally understood as the CEO's account to

shareholders of how the firm has performed (Patelli & Pedrini, 2014). Its prominence does not depend on any particular event or circumstance, which makes it the natural reference point for the analysis. Any additional attention devoted to a newly salient issue must be accommodated within a space that financial performance already occupies, so it serves as the baseline against which reallocation is assessed.

ESG is the domain in which the threat examined in this study arises. A decline in the composite Refinitiv Asset4 ESG score is an externally assigned and publicly visible assessment of performance in that domain (Gillan et al., 2021), placing ESG at the center of the situation the CEO must address. Unlike financial performance, which remains central under ordinary circumstances, ESG becomes especially important when recent performance in this domain has deteriorated. Legitimacy-based disclosure research supports this distinction. Firms with weaker environmental records tend to disclose more rather than less about environmental issues (Patten, 2002), and CEOs devote greater communicative attention to environmental matters when actual environmental performance is poor (Cong et al., 2014).

Diversity is an adjacent domain that competes for attention in its own right. It formally belongs to the social pillar of ESG, yet during the observation period its communicative salience was shaped by pressures such as activist scrutiny, investor demands, and public debate that were plausibly distinct from those affecting a firm's broader ESG standing. Firms with weaker diversity performance have been shown to rely on reassuring language in their diversity disclosures in order to manage stakeholder impressions, regardless of their actual diversity outcomes (Ben-Amar et al., 2022). Diversity is therefore socially grounded in a way that resembles ESG, but it is linked to a different set of pressures than the ESG signal examined here. This makes it a distinct destination for attention rather than simply part of the ESG response. It can either reinforce an ESG-focused response or compete with ESG for the same communicative space.

Innovation differs from the other topics because it is oriented toward the future rather than the present. It concerns what the firm aims to become, not only what it has recently been. This allows the CEO to frame the firm's standing in terms of future prospects rather than outcomes that have already been recorded and evaluated (Merkley, 2014). Innovation therefore represents a forward-looking alternative to an accountability-oriented response, making it the most relevant competing destination when a firm seeks to redirect attention away from a disappointing record.

The hypotheses examine how attention shifts between ESG and these competing topics when a firm's ESG standing is publicly called into question.

## **2.5 Hypotheses Development**

### **2.5.1 ESG Decline and the Reallocation of Attention toward ESG (H1)**

The previous sections argued that an ESG decline constitutes a publicly visible legitimacy threat and that the CEO letter is a limited space in which four topics compete for attention. Prior research has already shown that firms tend to communicate more about a domain when their performance in that domain is weak, particularly in the context of environmental disclosure (Patten, 2002; Cong et al., 2014). The remaining question is how this response occurs and how it affects the CEO's allocation of attention. The argument developed here is that two mechanisms can drive this response: one automatic and one deliberate.

The first mechanism operates automatically. Negative and threatening information tends to attract attention more readily than positive information, a tendency known as automatic vigilance (Pratto & John, 1991) that reflects a broader asymmetry in the way negative events dominate cognition (Baumeister et al., 2001). A public ESG decline has exactly these characteristics: it is salient, consequential, and imposed by an external assessment. If the content of the CEO letter reflects what occupies the CEO's attention at the time of writing, this mechanism predicts greater emphasis on ESG. No deliberate decision to foreground ESG is necessary, because the decline enters the executive's attention whether it is welcome or not.

The second mechanism is strategic. A CEO who engages in impression management has a strong incentive to address a damaged ESG standing directly in the shareholder letter. Doing so is relatively inexpensive. Bansal and Clelland (2004) show that firms with low environmental legitimacy, which otherwise face higher unsystematic stock-market risk, can reduce that risk by expressing commitment to the natural environment. Communicative emphasis can therefore produce benefits even when it is not accompanied by a corresponding change in behavior.

An alternative would be to redirect attention by emphasizing areas in which the firm performs well while giving less attention to controversial issues, a strategy the impression management literature describes as thematic manipulation (Brennan & Merkl-Davies, 2013). In the case of

an ESG decline, however, diversion is a comparatively weak response. The key difference lies in the nature of the signal. Weak earnings are raw figures that still require narrative framing, which leaves CEOs considerable discretion over how prominently they are presented. An ESG rating, by contrast, is an externally assigned and condensed assessment that stakeholders can observe before the CEO has an opportunity to respond (Gillan et al., 2021). Because the shareholder letter is limited in length, remaining silent about ESG after a public decline, or shifting conspicuously toward a more favorable topic, represents an active reallocation of attention away from ESG rather than a neutral omission. Such a response may therefore appear dismissive instead of responsive. As Suchman (1995) argues, restoring legitimacy requires an explanation that helps normalize the disruption, whereas responses perceived as insincere or as denying an obvious problem can further damage a firm's reputation. For a CEO acting strategically, the public and authoritative nature of an ESG decline therefore makes engagement more attractive than diversion in a way that is less likely for performance shortfalls that are less publicly visible.

If diversion rather than engagement were to govern the response, innovation would be its most plausible beneficiary. It is the only competing topic whose claim rests on the future rather than on the recorded present, which is exactly what a CEO seeking to redirect attention requires (Merkley, 2014). Financial performance cannot play this role because it is the topic from which any reallocation must draw rather than the topic toward which attention is redirected. Diversity is also an unsuitable diversion target because it speaks to the same legitimacy concern as ESG. Emphasizing diversity would therefore constitute a broader form of engagement rather than an escape from that concern. Because the CEO letter is limited in length, these alternatives imply two distinct empirical patterns. Under engagement, ESG emphasis rises while innovation emphasis does not. Under diversion, innovation emphasis rises while ESG emphasis remains flat or declines.

Both mechanisms converge on the same directional prediction for ESG, although one operates automatically and the other involves a deliberate choice constrained toward engagement. This study does not attempt to distinguish between them, since separating an involuntary attentional response from a calculated repair strategy would require evidence on internal managerial deliberation that the CEO letter does not provide.

Because the CEO letter is finite, an increase in ESG emphasis has a necessary counterpart in the topics with which it competes. Under engagement, this reallocation draws on the combined



space devoted to diversity, financial performance, and innovation, consistent with the relative measure used to test H1 rather than with a prediction about any single competing topic. Diversity is not assigned a directional prediction here because it captures a related but conceptually distinct legitimacy concern whose movement this study observes rather than predicts. The hypothesis is therefore stated in both absolute and relative terms, with the latter capturing the reallocation away from competing topics directly.

***H1:*** *Following a decline in ESG performance, CEOs increase the topical emphasis on ESG in their shareholder letters, both in absolute terms and as a share of the combined attention devoted to ESG, diversity, financial performance, and innovation.*

### **2.5.2 Ownership as a Boundary Condition (H2a and H2b)**

Hypothesis 1 assumes that CEOs respond to an ESG decline in the same way across firms. The CEO letter, however, is written for an audience, and the most consequential audience is the firm's owners. Owners are not homogeneous. They differ in their objectives, in the resources they devote to monitoring, and in the criteria they use to evaluate firm performance (Connelly et al., 2010). Following Collin et al. (2017) and Federo et al. (2020), this study distinguishes between two governance strategies that arise from these differences. Owners pursuing a financially oriented strategy, particularly institutional investors, hold diversified portfolios, develop general rather than firm-specific expertise, and evaluate firms using standardized indicators that facilitate comparisons across companies. Owners pursuing a firm-oriented strategy, including founding families, individual owners, and non-financial corporations with long-term stakes, concentrate their wealth in a particular firm, develop firm-specific knowledge, and assess performance against strategic goals developed for that firm.

The importance of this distinction for the CEO letter lies not in the difference in objectives itself, but in the difference in firm-specific knowledge that follows from it and the benchmark that this implies. Financially oriented owners hold diversified portfolios and value the ability to exit, so they invest less in firm-specific knowledge and rely more heavily on criteria that can be applied across a portfolio (Collin et al., 2017; Federo et al., 2020). An externally produced ESG rating is precisely such a portable and standardized criterion. Firm-oriented owners, by contrast, develop firm-specific knowledge through their concentrated and long-term involvement and evaluate the firm against criteria that reflect the trajectory of a particular business, for which a composite score assigned by a third party is a poor substitute (Collin et

al., 2017; Federo et al., 2020). The same ESG decline therefore carries a different weight depending on who is reading the letter, and this difference shapes how much of the letter's limited space a CEO has reason to devote to it.

For institutional investors, standardized ESG indicators have become part of the evaluative apparatus rather than a peripheral concern. They demand disclosure on environmental and social issues and pressure firms to improve their performance in these areas (Flammer et al., 2021). Higher institutional ownership is also associated with subsequent improvements in environmental and social performance, with engagement rather than divestment identified as the mechanism through which this influence operates (Dyck et al., 2019). Their reliance on publicly available information is evident in disclosure behavior more broadly. A meta-analysis covering 69 studies finds institutional ownership to be positively associated with the extent of voluntary disclosure (Khelif et al., 2017).

For a CEO facing an ESG decline, this has two implications. The decline speaks directly to the criterion that this audience uses to evaluate the firm, making it especially salient to the owners whose judgment matters most. At the same time, large institutional stakes cannot be sold quickly without cost (Edmans, 2014). These investors therefore cannot simply exit a firm whose standing has deteriorated and remain dependent on evidence that management recognizes the problem and intends to respond to it. The CEO letter is the place where such evidence is offered. As a result, both mechanisms underlying H1 become stronger: the decline becomes more salient when the audience most attuned to it owns a larger share of the firm, and public engagement becomes more valuable when that audience forms its judgment from public communication.

A possible objection is that large index-based institutions are too passive and too diversified to pay attention to any individual firm's CEO letter (Bebchuk & Hirst, 2019). Limited involvement in direct governance, however, does not imply indifference to public signals. It implies reliance on them, because a passive investment strategy rules out the firm-specific inquiry that could otherwise provide reassurance. Governance also does not require actual divestment. The credible threat of exit alone can discipline managers and create pressure to provide reassuring information, and this mechanism operates regardless of whether divestment itself is costly or constrained (Edmans, 2014).

For firm-oriented owners, the same ESG decline speaks to a criterion that plays a less important role in their evaluation of the firm. Given the firm-specific knowledge discussed above, these owners assess the firm primarily against long-term goals developed through close interaction with management, so a composite score assigned by a third party is not the main benchmark by which they judge whether the firm remains on course (Collin et al., 2017; Federo et al., 2020). This orientation is also reflected in their reporting behavior. The same meta-analysis finds ownership concentration to be negatively associated with the extent of voluntary disclosure, and the effect does not differ when ownership concentration is measured specifically as family ownership (Khlif et al., 2017). Family firms are also considerably less likely to comply with externally defined reporting standards such as the GRI guidelines, the Global Compact, or the Carbon Disclosure Project and instead orient their reporting toward characteristics that are specific to the firm (Campopiano & De Massis, 2015). This evidence points to a persistent orientation away from externally produced benchmarks rather than a reaction to any particular rating event. The present hypothesis extends this reasoning to the situation of an ESG decline. A CEO writing for this type of ownership base has correspondingly less reason to devote limited communicative space to a rating that the most important readers are not primarily using to evaluate the firm. This does not make the letter less important to these owners, but it does make this particular topic less pressing, thereby weakening both mechanisms underlying H1.

The strongest counterargument concerns founding families whose name is closely tied to the firm and who are often found to place particular value on corporate reputation (Deephouse & Jaskiewicz, 2013). The evidence on how such owners respond to environmental reputation threats, however, points toward substantive rather than rhetorical responses. Family-controlled firms in polluting industries reduce actual emissions in order to protect the family image (Berrone et al., 2010). Among firm-oriented owners, reputational sensitivity therefore tends to translate into pressure for real adjustment rather than for greater emphasis in the CEO letter.

Because ownership is expected to affect the strength of the response rather than its existence, both hypotheses apply to the reallocation defined for H1: ESG emphasis in absolute terms and ESG emphasis relative to the other focal topics.

***H2a:*** *The reallocation of communicative attention toward ESG following a decline in ESG performance is stronger when institutional ownership is high.*

*H2b: The reallocation of communicative attention toward ESG following a decline in ESG performance is weaker when firm-oriented ownership is high.*

### 3. Research Design

#### 3.1 Sample and Data

The sample consists of US-domiciled firms included in the Refinitiv Asset4 database and ranked by average total assets over the 2018–2022 period. Firms were selected in descending order of size, subject to two conditions: CEO shareholder letters had to be available, and data for all analysis variables had to be complete. A firm was replaced by the next-ranked firm if shareholder letters were unavailable for more than one or two years or if its Refinitiv Asset4 data were incomplete. Isolated single-year gaps did not lead to exclusion. The final sample therefore consists of the 150 highest-ranked firms that met these conditions rather than the 150 largest firms overall. Total assets is used as the size criterion because it is a standard and widely used measure of firm size in empirical corporate finance research (Dang, Li & Yang, 2018). The measure is averaged across all five years so that firm size is captured consistently over the observation window and is not distorted by year-specific fluctuations.

A multi-year panel is central to the research design for two reasons. Observing each firm over five years provides a sufficient number of observations for estimation and, more importantly, generates the within-firm variation over time on which the fixed-effects estimator relies (Wooldridge, 2010). Changes in a firm's ESG score can therefore be linked to changes in the composition of that firm's own CEO letter while holding constant all firm characteristics that do not vary over time. Firms whose ESG scores do not deteriorate in a given year remain in the sample and serve as a comparison group.

The focus on the largest firms is theoretically motivated. Large firms are substantially more visible to the public and face greater scrutiny from stakeholders than smaller firms, making them more exposed to legitimacy pressures when their ESG performance deteriorates (Patten, 2002; Suchman, 1995). Their CEO letters also tend to be longer and more content-rich, which is important for the textual analysis at the core of this study. The mechanism developed in Chapter 2, a legitimacy-driven reallocation of communicative attention following an ESG decline, depends on firms being subject to public scrutiny and on the CEO letter functioning

as an active channel of communication. The largest firms therefore provide the setting in which this mechanism is most likely to be observable.

The focus on US firms follows the same logic and is reinforced by two features of the US setting. First, CEO letters in the United States are largely unregulated in content and give executives considerable discretion over what they choose to emphasize, a precondition for studying topic choice as a strategic act (Brennan & Merkl-Davies, 2013). Second, the United States is a strongly individualistic culture in which firm outcomes are attributed to individual leaders to an unusually high degree. CEOs are therefore both more visible as the authors of the firm's narrative and more personally associated with its performance signals, making the discretionary choices reflected in the letter more likely to capture executive attention than in more collectivist settings (Hofstede, 2001; Hooghiemstra, 2010). The US context thus combines strong CEO influence over the shareholder letter with a clear channel through which an attentional response, if it exists, could become visible.

The unit of analysis is the firm-year. Each firm is observed once per year, so a firm covered across the full observation window contributes five observations, resulting in a panel of 150 firms and 750 firm-years. Because the independent variable is defined as a change relative to the previous year, 2018 serves as the base year and is excluded from the estimation. The estimation sample therefore covers 2019 to 2022 and contains 600 firm-year observations. Firm-level ESG scores, ownership data, and accounting variables are obtained from Refinitiv Asset4, while CEO shareholder letters are collected manually from corporate annual reports. A shareholder letter could not be obtained for ten firm-years. Six of these belong to the base year and therefore do not enter the estimation, leaving four observations lost because of missing letters. The final estimation sample consists of 596 firm-years for the absolute models and 582 for the relative measure, which is undefined when a letter contains none of the four focal topics (Section 3.3.1).

## 3.2 Keyword Library Construction

Communicative attention is measured through a dictionary-based content analysis of CEO letters, an approach with a well-established tradition in accounting and finance research (Loughran & McDonald, 2016). For each letter, the proportion of total words attributable to a given topic is calculated. The measure therefore captures communicative focus rather than sentiment: a term was included in a dictionary only if it clearly indicated that the topic was

being discussed, not because it expressed a positive or negative evaluation. Both an absolute and a relative version of the measure are used, as described in Section 3.3.1.

The measurement approach is not developed from scratch. It follows the dictionary-based content-analysis tradition established for financial disclosure by Loughran and McDonald (2011), whose keyword-in-context validation principle guides the construction of every dictionary used in this study. That approach is applied to the ESG and topic-attention setting in line with Baier, Berninger, and Kiesel (2020) and Ahn (2022). For more details on the use of AI tools in the preparation of this thesis, see Appendix D. The contribution at the measurement level therefore lies not in a new technique, but in the adaptation and validation of existing published dictionaries for the specific language of CEO letters and in their combination into four mutually exclusive topic dictionaries. The remainder of this section explains which published sources each dictionary builds on and how it was validated.

The keyword dictionaries are based on established published dictionaries wherever available and were subsequently validated and adapted for the specific context of CEO letters. For this purpose, a pilot sample of 20 CEO letters was compiled from firms in the estimation sample. The letters were drawn in equal proportions from firm-years with declining, stable, and improving ESG scores, classified according to the sign of the year-on-year change in the composite Refinitiv Asset4 ESG score (TRESGS), which is the same measure used to define the independent variable (Section 3.3.2). This three-way split is intended to preserve measurement neutrality. A dictionary developed only from letters written after an ESG decline would be calibrated to the vocabulary of that situation and could mechanically generate the predicted association. Drawing on all three states ensures that the dictionaries capture how the four topics are discussed more generally, which is also consistent with their application in the estimation sample, where declining, improving, and unchanged firm-years are all represented.

For most categories, a published dictionary served as the external starting point. For ESG, the environment and governance subcategories from Ahn (2022) were combined with the ESG dictionary of Baier et al. (2020). Because the Baier et al. (2020) dictionary was developed from 10-K reports and proxy statements, its legal and governance terminology had to be tested for suitability in the less formal language of CEO letters. For diversity, the diversity subcategory of Baier et al. (2020) served as the anchor and was supplemented with the machine-generated DEI dictionary of Demers, Wang, and Wu (2024), which contains 253 terms trained with word2vec on nearly 4,000 US corporate disclosures. For innovation, the research-and-

development glossary of Merkley (2014) provided the starting point, but it proved largely unsuitable for CEO letters because terms such as patent and R&D rarely appeared. The innovation dictionary was therefore developed primarily through an inductive process. For financial performance, no widely used topic-specific dictionary exists, so this dictionary was developed inductively from the pilot sample using the topic framework of Li (2010) and supplemented with financial-metric terms from the FinTextSim keyword list of Jehnen, Ordieres-Meré and Villalba-Díez (2025), which extends the word2vec-based dictionary of Fengler and Phan (2025). Throughout, the focus is on financial-metric vocabulary such as earnings, net income, and EBITDA rather than on emphasis markers such as record results, because the objective is to measure topic attention rather than tone.

Every candidate term was validated against the 20 pilot letters before being included in the final dictionaries, following the keyword-in-context approach developed by Loughran and McDonald (2011) and applied by Baier et al. (2020). All occurrences of each candidate term in the pilot letters were identified, and the surrounding context was examined to determine whether the term indicated the intended topic or represented a false positive. Final inclusion was determined by the author. Two rules guided this process. A term that appeared in the pilot was retained only if its usage clearly signaled the intended topic. Terms that were predominantly ambiguous or belonged conceptually to a different construct were excluded even when they occurred frequently. A term that did not appear in the pilot was retained only if it belonged unambiguously to the topic on conceptual grounds, since the absence of a term from 20 letters does not demonstrate that it is unsuitable for a corpus of 750 letters. Terms with a second, unrelated meaning were removed regardless of frequency.

The validation showed that the published dictionaries were only partly transferable. A substantial number of candidate terms had to be removed, particularly in the ESG category, because both source dictionaries were developed from 10-K filings and proxy statements. Governance terms such as board, policy, oversight, and transparency occur frequently in CEO letters but rarely indicate that governance is actually being discussed, and were therefore excluded. Similar exclusions were made when a term referred to business activity rather than the underlying construct, as with manufacturing, logistics, and supply chain in the financial-performance dictionary, or when a term had an established second meaning, as with disruption in the innovation dictionary and background in the diversity dictionary. In a small number of cases, a term was retained but its known false-positive context was excluded from the search

pattern. Examples include recycling excluding recycling capital, profits excluding not-for-profits, minority excluding minority interest, and black excluding the phrase in the black. A limited number of recent terms were added inductively, particularly acronyms such as DE&I and phrases such as net zero.

The final dictionaries contain 75 ESG terms, 59 diversity terms, 41 innovation terms, and 35 financial-performance terms. Applied to the 20 pilot letters, the four dictionaries jointly cover an average of 1.36 percent of total letter text, with values ranging from 0.30 to 4.78 percent across letters. Each term was assigned to exactly one category. All diversity terms were assigned exclusively to the diversity dictionary and removed from the ESG dictionary, so the four topic shares do not overlap and shifts in emphasis between ESG and diversity remain observable rather than being absorbed within a single category. In implementation, each dictionary is applied through regular-expression alternation: all terms within a category are combined into a single search pattern, with longer phrases taking precedence over shorter ones. Each occurrence is therefore counted only once, even when it satisfies multiple entries. For example, if a pattern contains both earnings per share and earnings, the phrase is counted once and the embedded word is not counted separately. The complete dictionaries are reported in Appendix A.

Two features of the measure should be kept in mind. First, a dictionary captures explicit topic vocabulary rather than every passage in which a topic is discussed. This limitation is most relevant for financial performance, where results may be discussed extensively using evaluative language that contains no financial metric term, whereas innovation vocabulary such as artificial intelligence tends to appear explicitly whenever the topic is addressed. The reported shares should therefore be interpreted as the relative prominence of explicit topic vocabulary, and comparisons with studies that code content thematically rather than lexically (Conaway & Wardrope, 2010) should take this difference into account. Second, the four dictionaries differ in length, ranging from 35 to 75 terms. This reflects differences in the lexical breadth of the topics and in the size of the underlying published dictionaries rather than differences in the way the dictionaries were constructed. Similar variation in dictionary size is common in dictionary-based research, including the category dictionaries of Baier et al. (2020), on which several of the dictionaries used here are based. Because each dictionary remains fixed across all firm-years, differences in dictionary length can affect the level of a topic share but not its within-



firm variation over time, which is the variation used to test the hypotheses. Cross-topic comparisons of levels should nevertheless be interpreted with this difference in mind.

### 3.3 Variables

#### 3.3.1 Dependent Variables

For each firm-year observation, the absolute topic share is calculated for each of the four topics as:

$$\text{Topic Share} = (\text{Topic Words} / \text{Total Words}) \times 100$$

This measure follows established practice in the attention-based view literature, where managerial attention is operationalized as the proportion of letter text devoted to a particular topic and normalized by total letter length to enable comparisons across letters (Eggers & Kaplan, 2009).

H1 contains two distinct claims, and each requires a different measure. The first is that ESG emphasis increases, which is captured directly by the absolute ESG share. The second is that any increase in ESG emphasis occurs at the expense of the other focal topics. The absolute measure cannot establish this on its own because all four topic shares are calculated relative to total letter length and could, in principle, increase simultaneously. The reallocation claim is therefore tested with a relative measure that expresses ESG emphasis as a share of the attention devoted to the four focal topics combined:

$$\text{Relative ESG Attention} = \text{ESG Topic Share} / (\text{ESG} + \text{Diversity} + \text{Financial Performance} + \text{Innovation Topic Shares})$$

The two measures differ in the quantity their denominators hold constant. The absolute share is calculated relative to the entire letter and therefore also reflects how much of the letter is devoted to the four focal topics rather than to greetings, macroeconomic commentary, and other non-focal content. The relative measure conditions on the total amount of focal-topic content and captures how that space is divided across the four topics. The two measures therefore differ whenever the combined coverage of the four topics varies across letters. Because this variation determines how much additional information the relative measure provides, both the combined coverage of the four topics and the correlation between the two ESG measures are reported in

the descriptive statistics (Section 4.1). The relative measure is retained as a separate test only if that correlation indicates that it contains information beyond the absolute measure.

The main analysis therefore uses two dependent variables, each estimated in a separate model rather than entered jointly: the ESG share and relative ESG attention, both transformed using  $\log(1+x)$ . This transformation is motivated by the skew documented in Section 4.1 and preserves firm-year observations with a value of zero. The untransformed ESG share and a cube-root transformation are reported as robustness checks (Section 5.2). ESG mention is modeled separately as a binary outcome because a substantial number of CEO letters contain no ESG term (Section 3.3.2 and Section 4.6).

### 3.3.2 Independent Variable

The independent variable captures deterioration in a firm's ESG performance. It is based on the year-on-year percentage change in the composite Refinitiv Asset4 ESG score (TRESGS):  $\Delta ESG_t = (ESG_t - ESG_{t-1}) / ESG_{t-1} \times 100$ .

Expressing the change in percentage terms rather than in score points makes firms with different baseline ESG scores comparable. By construction, the same point change represents a larger percentage change for a firm with a low initial score than for a firm with a high initial score, so the relative importance of a given change depends on the firm's starting level.

The raw percentage change is not used directly because it treats improvements and deteriorations as symmetric opposites. The mechanisms developed in Chapter 2 predict an asymmetric response: negative information is processed more strongly than positive information of comparable size (Pratto & John, 1991; Baumeister et al., 2001), and only a deterioration creates the legitimacy gap that motivates a response. The percentage change is therefore decomposed into two variables. ESG Decline takes the absolute value of

$\Delta ESG_t$  when the change is negative and zero otherwise, so it measures the magnitude of deterioration. ESG Improvement is defined symmetrically for positive changes. ESG Decline is the variable of interest and is used to test the hypotheses, while ESG Improvement is retained so that improving firm-years are not pooled with unchanged observations. Because ESG Decline is coded as a positive magnitude, a positive coefficient indicates that larger deteriorations are associated with greater ESG emphasis, which is the direction predicted by

H1. In the estimation sample, 231 of the 596 firm-year observations (38.8 percent) record an ESG decline.

Modeling the trigger as a directional event rather than as a symmetric continuous variable is also consistent with prior attention-based research, which examines changes in the composition of shareholder letters around discrete shifts in a firm's environment rather than around continuous variation (Cho & Hambrick, 2006). As a robustness check, the models are re-estimated using a binary indicator that equals one when the ESG score declined, allowing the analysis to test whether the results depend on the magnitude of the decline or on its occurrence alone.

### **3.3.3 Moderating Variables**

Ownership structure is the moderating variable and is measured using two variables that represent the contrasting governance strategies developed in Chapter 2 (Collin et al., 2017; Federo et al., 2020). Institutional ownership is measured as the percentage of shares held by investment companies (NOSHIC). Firm-oriented ownership is measured as the percentage of strategic holdings (NOSHST) that is not held by investment companies, calculated as NOSHST minus NOSHIC. This adjustment is required because the Refinitiv Asset4 strategic-holdings variable includes all closely held shares, including those held by institutional investors. Using NOSHST directly would therefore cause the two ownership measures to overlap instead of representing contrasting ownership structures. The adjusted measure captures shares held by corporations, employees, and other concentrated long-term owners and corresponds to the firm-oriented governance strategy. Firm-years for which the database reports no strategic holdings are coded as zero rather than missing because the underlying ownership categories are also reported as zero. NOSHIC captures holdings by investment companies specifically and is therefore a subset of total institutional ownership. The reported mean is accordingly lower than headline institutional ownership figures for large US firms.

### **3.3.4 Control Variables**

To isolate the effect of ESG performance changes on topic emphasis from potential confounding factors, all models include the following control variables.

Firm size is measured as the natural logarithm of total assets. Larger firms generally devote more attention to ESG because they are subject to greater public scrutiny, independent of changes in their ESG performance (Patten, 2002). The expected association is therefore positive.

Prior financial performance is measured as return on assets in the year preceding the CEO letter. Firms with weak financial results may adjust their narrative for reasons unrelated to ESG, for example by increasing forward-looking communication to offset disappointing performance (Merkley, 2014). Controlling for prior financial performance helps separate the ESG-specific trigger from broader financial pressure.

Prior ESG score level is the firm's ESG rating in the year before the letter. It is included because the amount of attention devoted to ESG depends not only on how the score has changed but also on how important ESG already is within the firm. Firms with high ESG scores have typically integrated ESG more deeply into their strategy, governance, and external positioning and are therefore more likely to discuss it routinely in their shareholder letters. Firms with lower ESG scores have less institutional reason to raise the topic. The expected association is therefore positive, with higher prior ESG scores predicting greater ESG attention. This expectation is consistent with the data, where prior ESG is the strongest and most robust predictor of whether ESG is mentioned at all (Section 4.6). Including this control also ensures that the coefficient on ESG Decline captures changes in ESG performance rather than differences in baseline ESG levels. Two firms experiencing the same decline but starting from very different ESG levels face different communicative situations, and without this control they would be treated as equivalent. The variable does not, and is not intended to, capture chronically low-performing firms, whose ESG scores by construction exhibit little year-on-year variation and therefore little measured decline. Its purpose is to hold the baseline level constant while ESG Decline captures movement in the score.

Financial leverage is measured as total debt divided by common equity. Leverage reflects a firm's financial risk profile and resource constraints, both of which have been linked to CSR-related resource allocation (Lys et al., 2015). These conditions may also influence corporate communication independently of ESG performance changes, so leverage is included as a control variable. Because firms with near-zero or negative common equity generate extreme leverage ratios, the variable enters the models in signed logarithmic form.

### 3.4 Methodology

The hypotheses are tested using linear panel regressions estimated on the firm-year panel, with the moderated specification presented at the end of this section. The choice of estimator, pooled OLS, random effects, or fixed effects, is not assumed here but established empirically through formal specification tests. Following standard practice in panel-data analysis (Wooldridge, 2010), these tests are reported near the beginning of the empirical chapter, following the descriptive and bivariate results (Section 4.3). Standard errors throughout are heteroskedasticity-consistent (HC3) and clustered at the firm level to account for correlation among repeated observations within the same firm.

H1 is tested by regressing ESG emphasis on ESG Decline, together with ESG Improvement and the control variables. ESG emphasis, the dependent variable, is measured in two ways and estimated in separate main models: the ESG share and relative ESG attention, both transformed using  $\log(1+x)$ . The transformation is motivated by the skew analysis reported in Section 4.1 and preserves observations with a value of zero. The untransformed ESG share is reported only as a robustness check (Section 5.2), where it confirms that the null result is not driven by the functional form. H1 is supported if the coefficient on ESG Decline is positive and statistically significant.

The moderating hypotheses are tested by adding an interaction between ESG Decline and one ownership measure to the absolute ESG-share model, with each ownership measure entered separately together with its main effect. A positive interaction with institutional ownership supports H2a, whereas a negative interaction with firm-oriented ownership supports H2b. The moderated model is:

$$ESG\ Share_{it} = \beta_0 + \beta_1 ESG\ Decline_{it} + \beta_2 ESG\ Improvement_{it} + \beta_3 Ownership_{it} + \beta_4 (ESG\ Decline_{it} \times Ownership_{it}) + \gamma' Controls_{it} + \alpha_i + \delta_t + \varepsilon_{it}$$

where  $\alpha_i$  and  $\delta_t$  denote firm and year fixed effects, Ownership refers to either institutional or firm-oriented ownership, and Controls includes firm size, return on assets, leverage, and the prior-year ESG score. The H1 models correspond to this specification without the ownership and interaction terms and are estimated separately for each measure of ESG emphasis.

Two additional specifications extend the analysis beyond the confirmatory tests and are reported as post-hoc analyses in Section 4.6. Because a substantial share of CEO letters contains no ESG term, the decision to mention ESG is modeled separately from the amount of attention devoted to it. This is estimated using a logistic regression of a binary ESG-mention indicator on ESG Decline and the control variables, implemented as a pooled logit with year fixed effects and firm-clustered standard errors. A conditional fixed-effects logit is not used because the incidental-parameters problem would bias its estimates in a panel with many firms and only a few years of observations per firm. The model is also estimated using the lagged ESG change from  $t-2$  to  $t-1$ , since the CEO letter is published before the Refinitiv Asset4 score for year  $t$  is finalized and the most recent completed ESG change available at the time of writing is therefore the change from the previous year.

A second post-hoc analysis examines whether ESG emphasis varies with the severity of a decline. Firm-years with declining ESG scores are divided into quartiles based on the size of the decline and compared directly, using the mildest quartile as the reference category while holding the control variables, year fixed effects, and clustering constant. Both analyses are exploratory and are presented as such.

Two variables are transformed. Firm size enters the models as the natural logarithm of total assets. Leverage contains extreme values generated by near-zero equity denominators and is therefore transformed using the signed logarithm  $\text{sign}(x) \times \log(1+|x|)$ , which is defined for the full sample and preserves the sign of the ratio. Return on assets is left untransformed because it can take negative values. The robustness of the results is examined in Chapter 5 with respect to these specification choices, the functional form of the dependent variable, the treatment of the leverage outlier, the exclusion of zero-mention letters, the separate inclusion of ESG Decline and ESG Improvement, and the timing of the ESG score change.

## 4. Results

This chapter reports the empirical analysis in the following order: the descriptive statistics (Section 4.1), the bivariate correlations (Section 4.2), the formal tests used to choose the estimator (Section 4.3), the confirmatory regressions for H1 and the ownership hypotheses

(Sections 4.4 and 4.5), and two post-hoc analyses that extend the main model (Section 4.6). All robustness checks are collected in Chapter 5.

## 4.1 Descriptive Statistics

**Table I.** Descriptive statistics for the estimation sample.

| Variable                    | N   | Mean    | SD        | Min         | Median | Max       | % Zero |
|-----------------------------|-----|---------|-----------|-------------|--------|-----------|--------|
| ESG share                   | 596 | 0.361   | 0.504     | 0.000       | 0.138  | 2.800     | 35.2   |
| Diversity share             | 596 | 0.194   | 0.277     | 0.000       | 0.066  | 1.621     | 45.0   |
| Financial performance share | 596 | 0.609   | 0.518     | 0.000       | 0.477  | 2.890     | 7.4    |
| Innovation share            | 596 | 0.252   | 0.417     | 0.000       | 0.107  | 4.329     | 32.6   |
| Relative ESG attention      | 582 | 0.203   | 0.218     | 0.000       | 0.143  | 1.000     | 33.7   |
| ESG Decline                 | 596 | 2.939   | 6.347     | 0.000       | 0.000  | 76.030    | 61.2   |
| ESG Improvement             | 596 | 8.821   | 15.103    | 0.000       | 2.972  | 195.892   | 38.9   |
| Institutional ownership     | 595 | 10.104  | 9.460     | 0.000       | 8.000  | 81.000    | 20.7   |
| Firm-oriented ownership     | 595 | 6.279   | 16.331    | 0.000       | 0.000  | 90.000    | 75.8   |
| Firm size (log assets)      | 596 | 16.725  | 1.041     | 14.287      | 16.501 | 21.104    | 0.0    |
| Return on assets            | 596 | 4.977   | 7.848     | -123.660    | 3.415  | 28.020    | 0.0    |
| Prior-year ESG score        | 596 | 50.528  | 19.302    | 8.780       | 47.790 | 93.430    | 0.0    |
| Leverage (raw)              | 596 | 113.967 | 1,094.471 | -24,142.640 | 68.755 | 4,751.900 | 0.0    |

*Notes:* Estimation sample of 596 firm-years (150 firms, 2019–2022); the base year 2018 is excluded because the analysis relies on year-on-year change. The topic shares and ESG-change measures are available for 596 firm-years; relative ESG attention is defined for 582, because letters that mention none of the four topics are undefined on this ratio; the ownership variables are available for 595. “% Zero” is the share of firm-years for which the variable equals zero. Ownership variables and the prior-year ESG score are in percentage points; firm size is the natural logarithm of total assets; leverage is reported in raw form here and enters the models through the signed logarithm (Section 3.4).

As Table I shows, the four topic shares differ markedly in how prominent they are in CEO communication. Financial performance is by far the most common topic, with the highest mean share (0.609) and median (0.477), and it is absent from only 7.4 percent of letters. The other three topics appear much less frequently and contain a large number of zero observations. ESG has a mean share of 0.361 but a median of only 0.138, and 35.2 percent of letters contain no ESG keyword at all. Diversity is absent from 45.0 percent of letters, and innovation from 32.6 percent. Relative ESG attention has a mean of 0.203, indicating that ESG accounts for roughly one fifth of the combined attention devoted to the four focal topics, and it is zero for about one third of the letters for which the measure is defined.

Two features of these distributions are particularly important for the analysis. First, the means of all four topic shares are well above their medians, indicating pronounced right-skew. The difference is largest for ESG (0.223) and is of similar magnitude for the other three topics (0.128 to 0.145). Because ordinary least squares gives disproportionate weight to the upper tail when the dependent variable is heavily right-skewed, both ESG-emphasis measures, the

absolute ESG share and relative ESG attention, are estimated in log form. The  $\log(1+x)$  transformation is used in both cases because it reduces skew while preserving firm-year observations with a value of zero (Section 3.4). Second, a substantial share of observations is concentrated at zero. For many firm-years, the relevant outcome is therefore not how much a CEO writes about ESG but whether ESG is mentioned at all. A continuous-share model cannot fully capture this all-or-nothing feature, which motivates the logistic specification reported in Section 4.6.

The independent variable shows a similar pattern. ESG Decline is zero in 61.2 percent of firm-years, meaning that only 38.8 percent of observations record a year-on-year deterioration. Across all firm-years, the average decline is 2.939 percent and the maximum decline is 76.0 percent. ESG improvements are both more common and larger, with a mean of 8.821, zeros in 38.9 percent of firm-years, and a maximum of nearly 196 percent, reflecting the general upward trend in ESG scores during the observation period. An ESG decline is therefore the less common outcome, and the empirical analysis focuses on a relatively infrequent, though not rare, event.

The two ownership measures differ sharply. Institutional ownership averages 10.10 percent and is present in most firm-years, with only 20.7 percent of observations recording no institutional block. Firm-oriented ownership is zero in 75.8 percent of firm-years by construction because most firms in the sample have no firm-oriented block larger than institutional ownership. When such holdings are present, however, they take large and highly variable values ( $SD = 16.33$ ). This concentration in a relatively small subset of firms limits the variation available to identify the moderating effect of firm-oriented ownership, which is relevant for the interpretation of H2b (Section 4.5).

The control variables are consistent with a sample of large, established firms. Firm size (log assets) averages 16.725, with no missing values, while average profitability is modest ( $ROA = 4.977$ ). The minimum ROA reflects a single loss-making firm-year ( $-123.66$ ). Prior-year ESG scores range from 8.78 to 93.43, covering almost the full Refinitiv Asset4 range and showing substantial variation in baseline ESG standing. Leverage has an extreme minimum ( $-24,142.64$ ) and an inflated standard deviation. This is driven by two firms, GoDaddy and Home Depot, whose near-zero or negative common equity following large share repurchases produces extreme debt-to-equity ratios rather than data errors. To prevent these observations from dominating the estimation while retaining their sign and keeping the full sample, leverage



enters the models through the signed logarithm. Winsorising at the 1st and 99th percentiles is examined as an alternative and leads to the same conclusion (Section 5.1).

## 4.2 Correlation Matrix

Table II reports the pairwise correlations among all variables. Multicollinearity is not a concern in the estimated models. A common convention treats an absolute correlation above 0.70 as problematic (Dormann et al., 2013), but no pair of predictors entered jointly into a regression comes close to this threshold. The largest correlation is 0.38, between firm size and the prior-year ESG score. The two ESG measures are strongly correlated ( $r = 0.85$ ), which is expected because relative ESG attention is constructed from the ESG share. They are therefore never included in the same model.

**Table II.** Pairwise correlation matrix (N = 581-600).

| Variable             | 1       | 2       | 3        | 4        | 5        | 6        | 7        | 8        | 9        | 10      | 11      | 12    | 13   |
|----------------------|---------|---------|----------|----------|----------|----------|----------|----------|----------|---------|---------|-------|------|
| 1 ESG share          | 1.00    |         |          |          |          |          |          |          |          |         |         |       |      |
| 2 Diversity share    | 0.23*** | 1.00    |          |          |          |          |          |          |          |         |         |       |      |
| 3 Fin. perf. share   | -0.02   | -0.12** | 1.00     |          |          |          |          |          |          |         |         |       |      |
| 4 Innovation share   | 0.12**  | 0.03    | -0.17*** | 1.00     |          |          |          |          |          |         |         |       |      |
| 5 Rel. ESG attention | 0.85*** | 0.12**  | -0.20*** | -0.02    | 1.00     |          |          |          |          |         |         |       |      |
| 6 ESG decline        | -0.07   | -0.05   | -0.03    | -0.01    | -0.04    | 1.00     |          |          |          |         |         |       |      |
| 7 ESG improvement    | -0.12** | -0.07   | 0.05     | -0.01    | -0.14*** | -0.27*** | 1.00     |          |          |         |         |       |      |
| 8 Institutional own. | -0.08   | -0.04   | 0.00     | -0.17*** | -0.08    | 0.10*    | 0.03     | 1.00     |          |         |         |       |      |
| 9 Firm-oriented own. | -0.05   | -0.04   | -0.09*   | -0.02    | 0.00     | -0.02    | -0.01    | -0.28*** | 1.00     |         |         |       |      |
| 10 Firm size         | 0.10*   | 0.16*** | -0.17*** | 0.11**   | 0.13**   | -0.04    | -0.13**  | -0.25*** | 0.02     | 1.00    |         |       |      |
| 11 ROA               | 0.09*   | -0.03   | 0.04     | 0.20***  | 0.07     | -0.05    | -0.06    | -0.15*** | -0.06    | 0.07    | 1.00    |       |      |
| 12 Prior ESG score   | 0.42*** | 0.19*** | -0.10*   | 0.24***  | 0.40***  | -0.07    | -0.35*** | -0.25*** | -0.28*** | 0.38*** | 0.26*** | 1.00  |      |
| 13 Leverage          | 0.00    | -0.05   | 0.01     | -0.04    | 0.00     | -0.00    | -0.02    | -0.01    | 0.01     | 0.05    | 0.04    | -0.00 | 1.00 |

Notes: Pairwise correlations; N ranges from 581 to 600 across pairs, reflecting differing missing-data patterns across variables. The upper bound of 600 arises for pairs involving only Refinitiv Asset4-based variables (e.g., ESG Decline, ownership, leverage), which are available for the full 2019–2022 panel; the four topic shares and relative ESG attention are limited to the 596 and 582 firm-years, respectively, for which a CEO letter was available (Section 3.1). The lower bound of 581 arises where the missing observations of two paired variables do not fully overlap. \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ . Column numbers correspond to the row labels. Leverage is shown in raw form; it enters the models as the signed logarithm.

Two aspects of the matrix inform the analysis that follows. First, the four topic shares are only modestly correlated, and the signs are not uniformly negative. ESG and diversity are weakly positively related ( $r = 0.23$ ,  $p < 0.001$ ), which supports treating them as separate topics rather than as one ESG construct. Financial performance is negatively related to both diversity ( $r = -0.12$ ) and innovation ( $r = -0.17$ ). The competing-attention idea in Section 2.4 would imply negative correlations among all focal topics. The raw correlations show this only for some pairs. These are, however, correlations without controls and without the fixed-effects structure of the main models. They also neither theorize nor test a specific pairwise trade-off. They are therefore treated here only as a weak, preliminary indication and not as a test of the competing-attention claim. The regression evidence in Section 4.4 provides the basis for conclusions on reallocation.

Second, the two change measures relate differently to ESG emphasis. ESG decline is not significantly correlated with the ESG share ( $r = -0.07$ ) or with relative ESG attention ( $r = -0.04$ ). ESG improvement is significantly negatively correlated with both ( $r = -0.12$ ,  $p < 0.01$ ;  $r = -0.14$ ,  $p < 0.001$ ). At the descriptive level, it is a prior improvement rather than a prior decline that is associated with lower ESG communication. This distinction is examined directly in the regressions in Sections 4.4 and 4.6.

### 4.3 Model Specification Tests

The choice of estimator is established through formal tests before estimating the hypotheses. The first question is whether the panel structure matters at all, that is, whether pooled OLS is adequate or firm-specific effects are present. An F-test for individual effects decisively rejects the pooled model ( $F(149, 440) = 6.33$ ,  $p < 0.001$ ; Table III). Unobserved firm heterogeneity is therefore present, so a panel estimator is required. The second question is whether these effects should be modeled as fixed or random. A Hausman test on the absolute ESG-share model rejects the random-effects specification ( $\chi^2(6) = 14.89$ ,  $p = 0.021$ ). This indicates that the random-effects assumption of orthogonality between the firm effects and the regressors does not hold, making fixed effects the consistent choice. Fixed effects are therefore used for the main models. Because firm fixed effects absorb all stable unobserved differences across firms, including time-invariant industry membership, identification rests on within firm variation over time, and no separate industry controls are required. Year fixed effects are included throughout

to capture common shocks, including the general rise in ESG communication over the sample period.

**Table III.** Specification tests for the choice of estimator.

| Test                                             | Statistic            | p         | Conclusion                                        |
|--------------------------------------------------|----------------------|-----------|---------------------------------------------------|
| F-test: pooled OLS vs. fixed effects             | $F(149, 440) = 6.33$ | $< 0.001$ | Individual effects present; panel model required  |
| Hausman: fixed vs. random effects (ESG share)    | $\chi^2(6) = 14.89$  | 0.021     | Fixed effects preferred                           |
| Hausman: fixed vs. random effects (relative ESG) | $\chi^2(6) = 6.56$   | 0.364     | Random effects consistent; fixed effects retained |

*Notes: F-test for individual effects and Hausman tests comparing fixed- and random-effects estimators. For the relative-ESG model the Hausman test does not reject random effects, but fixed effects are retained so that all models share one estimator; the random-effects estimates yield the same substantive conclusion (Table IV).*

For completeness, and because the Hausman test does not reject random effects for the relative-ESG model ( $\chi^2(6) = 6.56$ ,  $p = 0.364$ ), this model was also estimated using random effects. A non-significant Hausman test would in principle favor random effects because the random-effects estimator is more efficient and exploits both within- and between-firm variation. However, adopting random effects requires controlling for firm-invariant characteristics, most importantly industry affiliation, to satisfy the orthogonality assumption. Industry classification is not available in the current dataset, so random effects cannot be implemented without risking omitted-variable bias from unobserved, time-invariant firm heterogeneity. Fixed effects are therefore retained for all models, including the relative-ESG specification, as the conservative choice that eliminates this source of bias. The random-effects coefficient on ESG Decline for relative ESG attention is  $-0.0001$  ( $p = 0.939$ ), confirming that the substantive conclusion is unaffected by the choice of estimator.

#### 4.4 ESG Decline and ESG Attention (H1)

H1 predicts that a decline in a firm's ESG score leads its CEO to devote more of the letter to ESG. It is tested by regressing ESG emphasis, the dependent variable, on ESG Decline, controlling for ESG Improvement and the full set of controls, with firm and year fixed effects and firm-clustered standard errors. The main test uses two dependent variables, each estimated in a separate model: the ESG share and relative ESG attention. Both are entered in log form to address the pronounced right-skew documented in Section 4.1. The  $\log(1+x)$  transformation reduces skew while retaining firm-years with a value of zero. The untransformed ESG share is not reported as a main model but appears as a robustness check (Table X), where it confirms that the null result is not an artifact of the transformation. Support for H1 requires a positive

and significant coefficient on ESG Decline. Table IV reports the full models, including the coefficient, standard error, and two-sided p-value for every variable.

**Table IV.** Fixed-effects regressions for H1.

| Variable                | M1: log ESG share | p     | M2: log relative ESG | p     |
|-------------------------|-------------------|-------|----------------------|-------|
| ESG Decline             | 0.0000 (0.0012)   | 0.977 | 0.0006 (0.0007)      | 0.361 |
| ESG Improvement         | -0.0003 (0.0006)  | 0.607 | -0.0003 (0.0004)     | 0.455 |
| Firm size               | -0.0206 (0.0462)  | 0.656 | -0.0160 (0.0255)     | 0.532 |
| Return on assets        | -0.0004 (0.0011)  | 0.747 | -0.0013 (0.0006)     | 0.025 |
| Prior-year ESG score    | 0.0000 (0.0021)   | 0.993 | -0.0008 (0.0013)     | 0.530 |
| Leverage (signed log)   | -0.0134 (0.0055)  | 0.015 | -0.0097 (0.0044)     | 0.029 |
| Firm fixed effects      | Yes               |       | Yes                  |       |
| Year fixed effects      | Yes               |       | Yes                  |       |
| Observations            | 596               |       | 582                  |       |
| Firms                   | 150               |       | 150                  |       |
| R <sup>2</sup> (within) | 0.014             |       | 0.026                |       |

*Notes:* Two-way (firm and year) fixed-effects regressions; firm-clustered HC3 standard errors in parentheses; p is the two-sided p-value reported in the adjacent column. Coefficients rounded to four decimals; p is the two-sided p-value. Both dependent variables enter in log form using the  $\log(1+x)$  transformation, which addresses the right-skew while preserving the firm-years that record a zero (Section 4.1). The untransformed ESG share is reported as a robustness check (Table X). Support for H1 requires a positive and significant coefficient on ESG Decline.

No statistically significant association is detected between ESG Decline and ESG communication in either model. The coefficient is essentially zero for the log ESG share (0.0000,  $p = 0.977$ ) and small, positive, but insignificant for log relative ESG attention (0.0006,  $p = 0.361$ ). H1 is not supported in either absolute or relative terms. ESG Improvement is also insignificant in both contemporaneous models ( $p \geq 0.455$ ). This becomes relevant in Section 4.6, where the timing of the score relative to the letter is addressed.

The controls behave in an informative way. Leverage in signed-logarithmic form is a consistently significant predictor and enters negatively in both models ( $p < 0.05$ ). Firms with higher leverage therefore devote less of the letter to ESG. Return on assets is significant in the log relative-ESG model ( $-0.0013$ ,  $p = 0.025$ ), where more profitable firms give relatively less space to ESG, but it is insignificant in the absolute model. Firm size and the prior-year ESG score are insignificant in both models. Their insignificance here, despite being established predictors of ESG communication in cross-sectional work, is expected rather than anomalous. Both variables are largely stable within a firm over a five-year window, so the firm fixed effects absorb most of their variation and leave little for the coefficients to capture. The contrast is visible in Section 4.6, where the prior ESG level, freed from the firm effects in the pooled logit, becomes the strongest predictor of whether ESG is mentioned at all. The within R<sup>2</sup> is low in both models (1.4 and 2.6 percent), indicating that the change measures explain little of the

within-firm variation in ESG communication. The models thus provide no support for the hypothesized reallocation.

#### 4.5 The Moderating Role of Ownership (H2a and H2b)

The ownership hypotheses propose that the H1 relationship is stronger where institutional ownership is high (H2a) and weaker where firm-oriented ownership is high (H2b). Each hypothesis is tested by adding an interaction between ESG Decline and one ownership measure to the log ESG-share model, together with the corresponding main effect. Table V reports both models in full.

**Table V.** Fixed-effects regressions for H2a and H2b.

| Variable                           | M3 ( $\times$ Institutional) | p     | M4 ( $\times$ Firm-oriented) | p     |
|------------------------------------|------------------------------|-------|------------------------------|-------|
| ESG Decline                        | −0.0006 (0.0019)             | 0.742 | 0.0005 (0.0017)              | 0.767 |
| ESG Improvement                    | −0.0003 (0.0006)             | 0.633 | −0.0003 (0.0006)             | 0.625 |
| Institutional ownership            | −0.0000 (0.0029)             | 0.996 | —                            | —     |
| Firm-oriented ownership            | —                            | —     | 0.0011 (0.0033)              | 0.732 |
| ESG Decline $\times$ Institutional | 0.0001 (0.0001)              | 0.616 | —                            | —     |
| ESG Decline $\times$ Firm-oriented | —                            | —     | −0.0001 (0.0001)             | 0.534 |
| Firm size                          | −0.0211 (0.0462)             | 0.648 | −0.0212 (0.0461)             | 0.646 |
| Return on assets                   | −0.0003 (0.0011)             | 0.752 | −0.0003 (0.0011)             | 0.752 |
| Prior-year ESG score               | 0.0001 (0.0021)              | 0.960 | 0.0000 (0.0021)              | 0.990 |
| Leverage (signed log)              | −0.0135 (0.0054)             | 0.014 | −0.0136 (0.0053)             | 0.011 |
| Firm & year fixed effects          | Yes                          |       | Yes                          |       |
| Observations                       | 595                          |       | 595                          |       |
| Firms                              | 150                          |       | 150                          |       |
| R <sup>2</sup> (within)            | 0.014                        |       | 0.014                        |       |

*Notes:* Two-way fixed-effects regressions of the log ESG share; firm-clustered HC3 standard errors in parentheses. *p* is the two-sided *p*-value. H2a is supported by a positive and significant interaction with institutional ownership; H2b by a negative and significant interaction with firm-oriented ownership.

Neither interaction is statistically significant. The interaction with institutional ownership is 0.0001 ( $p = 0.616$ ), while the interaction with firm-oriented ownership is −0.0001 ( $p = 0.534$ ). Although both signs are in the predicted direction, an insignificant coefficient does not provide evidence that either type of owner strengthens or weakens the relationship. H2a and H2b are therefore not supported. The ownership main effects are also insignificant. As in the H1 models, leverage in signed-logarithmic form is the only consistently significant predictor, entering negatively in both models ( $p < 0.05$ ). One caveat, noted in Section 4.1, bears on H2b. Firm-oriented ownership is zero for 75.8 percent of firm-years, so its interaction is identified from a minority of firms, which limits the power of the test. More fundamentally, because H1 detects no baseline reallocation, there is no main relationship for ownership to condition.

## 4.6 Post-hoc Analysis: The Decision to Mention ESG

The confirmatory tests detect no association between an ESG decline and greater ESG emphasis. This section extends the main model in two exploratory directions suggested by the structure of the data. It does not aim to verify the earlier results. Both extensions are motivated by the all-or-nothing character of the dependent variable noted in Section 4.1. For a third of the letters, the relevant question is not how much a CEO writes about ESG but whether the topic appears at all. The first extension models this binary decision. The second compares firms with mild and severe declines directly. Both analyses are exploratory and were carried out after the confirmatory tests. They are therefore reported as patterns observed in the data that run counter to H1, rather than as hypotheses that were specified in advance.

Table VI reports the mention model, a pooled logistic regression of a binary indicator equal to one if the letter mentions ESG on ESG Decline and the controls. Two versions are estimated. M5 uses the contemporaneous change. M6 uses the change lagged from  $t-2$  to  $t-1$ , which represents the movement a CEO could actually observe when writing, since the score for the current fiscal year is finalized only after the letter is published.

**Table VI.** Logistic regression of the decision to mention ESG.

| Variable                       | M5: contemporaneous | p       | M6: lagged ( $t-2 \rightarrow t-1$ ) | p       |
|--------------------------------|---------------------|---------|--------------------------------------|---------|
| ESG Decline / lagged decline   | −0.0409 (0.0210)    | 0.052   | −0.0162 (0.0209)                     | 0.437   |
| ESG Improvement / lagged impr. | −0.0052 (0.0063)    | 0.410   | −0.0286 (0.0083)                     | < 0.001 |
| Firm size                      | 0.2166 (0.1460)     | 0.138   | 0.2200 (0.1724)                      | 0.202   |
| Return on assets               | 0.0074 (0.0155)     | 0.633   | −0.0019 (0.0134)                     | 0.889   |
| Prior-year ESG score           | 0.0464 (0.0091)     | < 0.001 | 0.0557 (0.0107)                      | < 0.001 |
| Leverage (signed log)          | 0.0088 (0.0754)     | 0.907   | −0.0020 (0.0876)                     | 0.982   |
| Year dummies                   | Yes                 |         | Yes                                  |         |
| Firm fixed effects             | No (pooled)         |         | No (pooled)                          |         |
| Observations                   | 596                 |         | 447                                  |         |
| Firms (clusters)               | 150                 |         | 150                                  |         |
| AME of decline measure         | −0.0093             |         | −0.0036                              |         |

*Notes:* Pooled logistic regressions with firm-clustered standard errors in parentheses; a conditional fixed-effects logit is not estimated because of the incidental-parameters problem. The dependent variable equals one if the letter mentions ESG. M5 uses the contemporaneous change; M6 the lagged change ( $t-2$  to  $t-1$ ), the information a CEO could observe when writing. “AME” is the average marginal effect of the respective decline measure, on the probability scale.  $p$  is the two-sided  $p$ -value.

In the contemporaneous model (M5), ESG Decline carries a negative coefficient of marginal significance (−0.0409,  $p = 0.052$ ), with an average marginal effect of −0.93 percentage points on the probability of mentioning ESG per one-percent unit of decline. In the timing-corrected model (M6), which uses the change a CEO could actually observe, the coefficient is insignificant (−0.0162,  $p = 0.437$ ). ESG Improvement, however, becomes strongly significant

and negative ( $-0.0286$ ,  $p < 0.001$ ): firms whose ESG score improved in the observable window are significantly less likely to mention ESG. In both models, the direction is opposite to H1. Movement in the score is associated with less ESG communication, not more.

The prior-year ESG score is the strongest predictor in both models ( $p < 0.001$ ), confirming the positive direction anticipated in Section 3.3.4. Firms with a higher standing are markedly more likely to raise ESG at all. Leverage, the one control that is consistently significant in the fixed-effects models of Table IV, is not significant here ( $p = 0.907$  and  $p = 0.982$ ). The mention decision and the degree of emphasis therefore appear to respond to different firm characteristics. The pooled cross-sectional specification of the logit also lacks firm fixed effects and is not directly comparable to the panel models of Table IV. The two models differ in both the change measure and the sample, so they should not be read as directly comparable. They are reported together to show that the negative sign on decline does not survive the timing correction, whereas the negative sign on improvement strengthens.

The second step asks whether the severity of a decline matters. The declining firm-years are ordered by magnitude and split into quartiles, from the mildest (Q1) to the most severe (Q4). Descriptively, ESG communication falls as the decline deepens (Table VII). The mean ESG share drops from 0.436 in Q1 to 0.232 in Q4, while the share of letters omitting ESG rises from 20.7 to 60.3 percent.

**Table VII.** Descriptive ESG attention across quartiles of decline magnitude.

| Quartile (decline range) | N  | Mean ESG share | Median ESG share | % with no ESG mention |
|--------------------------|----|----------------|------------------|-----------------------|
| Q1 (0.03–2.31%)          | 58 | 0.436          | 0.296            | 20.7                  |
| Q2 (2.31–5.21%)          | 58 | 0.416          | 0.213            | 24.1                  |
| Q3 (5.21–10.46%)         | 57 | 0.314          | 0.087            | 36.8                  |
| Q4 (10.46–76.03%)        | 58 | 0.232          | 0.000            | 60.3                  |

*Notes: The 231 declining firm-years, ordered by the magnitude of the decline and split into quartiles. The table is descriptive and holds no other factor constant; the multivariate test follows in Table VIII.*

A descriptive comparison cannot rule out other explanations, so the pattern is tested directly, with Q1 as the reference category and the full set of controls, year dummies, and firm-clustered standard errors, for both dependent variables. Table VIII reports the full models.



**Table VIII.** Direct test of quartile differences (declining firm-years; Q1 = reference).

| Variable                         | ESG share (OLS)  | p       | Mentions ESG (logit) | p     |
|----------------------------------|------------------|---------|----------------------|-------|
| Q2 vs. Q1                        | −0.0554 (0.0865) | 0.523   | −0.3165 (0.5083)     | 0.533 |
| Q3 vs. Q1                        | −0.0817 (0.0771) | 0.290   | −0.7331 (0.4639)     | 0.114 |
| Q4 vs. Q1                        | −0.0455 (0.0701) | 0.517   | −1.3375 (0.4537)     | 0.003 |
| Firm size                        | −0.0317 (0.0264) | 0.231   | 0.3904 (0.1942)      | 0.044 |
| Return on assets                 | −0.0012 (0.0057) | 0.840   | 0.0278 (0.0357)      | 0.436 |
| Prior-year ESG score             | 0.0096 (0.0018)  | < 0.001 | 0.0295 (0.0105)      | 0.005 |
| Leverage (signed log)            | 0.0230 (0.0123)  | 0.063   | −0.0543 (0.1133)     | 0.631 |
| Observations                     | 231              |         | 231                  |       |
| R <sup>2</sup> / AME (Q4 vs. Q1) | 0.281            |         | −0.306               |       |

*Notes: Declining firm-years only (N = 231; quartiles of 58, 58, 57, 58). Both models include ESG Improvement (not shown) and the controls, year dummies, and firm-clustered standard errors. OLS dependent variable: ESG share; logistic dependent variable: the mention indicator. The final row reports the OLS R<sup>2</sup> and the logistic average marginal effect for Q4 versus Q1 (−0.306, i.e. −30.6 percentage points). p is the two-sided p-value.*

The two outcomes give different answers, and the difference is informative. For the amount of ESG communication, measured by the OLS on the share, no quartile differs significantly from Q1. The level differences in Table VII therefore do not survive the controls. For the decision to mention ESG at all, the most severe quartile differs sharply from the mildest. Firms in Q4 are significantly less likely to mention ESG than firms in Q1 (−1.3375,  $p = 0.003$ ), a gap of roughly 31 percentage points in the probability of raising the topic. The pattern is therefore concentrated on the extensive margin, whether ESG appears rather than how much is written. This is consistent with the concentration of the outcome at zero. It is also non-linear, emerging only at the most severe declines. The quartile regression contrasts the severe quartiles with the mildest while holding the controls constant. It therefore provides a stronger test of whether emphasis differs by severity than an unconditional comparison of subsample means. Among the controls, the prior-year ESG score is significant in both models and firm size is significant in the logit ( $p = 0.044$ ). The direction is consistent with expectations in each case.

Three qualifications limit the interpretation to an exploratory finding. First, this is a contemporaneous association. The quartiles are defined on the same-year change, so the relationship is reported as an association rather than as a response to an observed decline. Second, the pattern does not survive the timing correction. Under the lagged model (M6), the decline coefficient is insignificant, so the pattern cannot be attributed to CEOs reacting to a deterioration they had seen. Third, its direction cannot be established. Firms that already say little about ESG may be those whose ratings later fall. The pattern is therefore an exploratory association, not causal evidence.

Overall, the post-hoc results run counter to H1. CEOs are not more likely to emphasize ESG when their score has declined. If anything, the deepest declines are associated with a lower probability of raising ESG at all, while under the observable-timing model it is prior improvement that predicts silence. Neither pattern is anticipated by the framework of Chapter 2, and both are carried into the discussion as unexpected findings.

## 5. Robustness Checks

The results in Chapter 4 rest on a set of modeling choices, including how the extreme leverage value is treated, the functional form of the dependent variable, whether firms that never mention ESG are included, whether the two change measures are entered together, and the timing of the score change. This chapter verifies that the central conclusion, namely the absence of a significant association between an ESG decline and ESG communication, does not depend on any of these choices. The binary mention model, the timing-corrected model, and the quartile test already reported in Section 4.6 provide further robustness evidence and are not repeated here.

### 5.1 Alternative Treatment of Leverage

Leverage contains an extreme value produced by two firms with near-zero or negative common equity (Section 4.1). The main models use the signed logarithm, which is defined for the full sample and reduces the influence of the outlier while preserving its sign. To confirm that this treatment does not drive the results, the H1 model is re-estimated with leverage winsorised at the 1st and 99th percentiles and with leverage in raw, untransformed form.

**Table IX.** ESG Decline coefficient under alternative treatments of leverage.

| Leverage treatment                  | ESG Decline coef (SE) | p     | Leverage p | N   |
|-------------------------------------|-----------------------|-------|------------|-----|
| Signed logarithm (untransformed DV) | −0.0001 (0.0021)      | 0.961 | 0.027      | 596 |
| Winsorised (1st/99th percentile)    | −0.0002 (0.0022)      | 0.937 | 0.052      | 596 |
| Raw (untransformed)                 | −0.0000 (0.0021)      | 0.998 | 0.314      | 596 |

*Notes:* Each row re-estimates the untransformed absolute ESG-share model; the dependent variable is the raw ESG share in all three rows; the corresponding log-ESG-share (M1) results are reported in Table IV. Each row varies only the treatment of leverage; all other controls, fixed effects, and clustering are identical. “Leverage p” is the two-sided p-value of the leverage control under each treatment.

The coefficient on ESG Decline is essentially zero and insignificant under all three treatments (Table IX), so the null result for H1 is not an artifact of the signed-logarithm transformation. The leverage control itself is significant only under the signed logarithm and marginal under winsorising. This is expected because the transformation determines how the outlier enters the model, but it affects only the control and not the variable of interest. The signed logarithm is retained because it is defined for every firm and requires no arbitrary cut-off. Winsorising, the more standard treatment of such outliers, leads to the same conclusion for ESG Decline.

## 5.2 Alternative Functional Form and Sample

The absolute ESG share enters the main models in log form to address its right-skew and concentration at zero (Section 4.1). Three specifications examine whether the null depends on the distribution of the outcome. The first uses the untransformed share, reported here rather than as a main model in line with the preference for the log specification. The second uses a cube-root transformation, which reduces skew and is defined at zero. The third restricts the sample to firms that mention ESG at all, thereby removing the mass of zeros entirely.

**Table X.** ESG Decline coefficient under alternative functional form and sample.

| Specification                   | ESG Decline coef (SE) | p     | N   |
|---------------------------------|-----------------------|-------|-----|
| Untransformed ESG share         | −0.0001 (0.0021)      | 0.961 | 596 |
| Cube-root ESG share             | −0.0011 (0.0018)      | 0.542 | 596 |
| Subsample: firms mentioning ESG | −0.0009 (0.0035)      | 0.787 | 386 |

*Notes: All specifications re-estimate the H1 model with the signed-logarithm leverage control and the full set of controls, fixed effects, and firm-clustered standard errors. The untransformed row uses the ESG share in levels; the cube-root row applies  $x^{(1/3)}$  to the ESG share; the subsample row restricts estimation to firm-years with a non-zero ESG share (386 firm-years, 134 firms).*

None of the three specifications alters the conclusion. The coefficient on ESG Decline remains small and insignificant under all three specifications (Table X). Together with the log-transformed model in Table IV, these results show that the failure to support H1 is not driven by the skew of the dependent variable or by the large share of letters that omit ESG entirely.

## 5.3 Alternative Timing of the ESG Score Change

The main models relate each letter to the ESG-score change of the same fiscal year, but that score is finalized only after the letter is written. The most recent change a CEO can actually observe is therefore the one completed in the prior year. To confirm that the null does not stem from this timing mismatch, the H1 models are re-estimated using the score change from the

t-2-to-t-1 window, which is observable when the letter is written, alongside the contemporaneous model on the same reduced sample.

**Table XI.** ESG Decline coefficient under alternative timing of the score change.

| Specification                                | Decline coef (SE) | p     | N   |
|----------------------------------------------|-------------------|-------|-----|
| ESG share — contemporaneous (reduced sample) | -0.0002 (0.0027)  | 0.937 | 447 |
| ESG share — lagged (t-2 to t-1)              | 0.0035 (0.0038)   | 0.352 | 447 |
| Relative ESG — lagged (t-2 to t-1)           | 0.0033 (0.0023)   | 0.154 | 436 |

*Notes: Each row re-estimates an H1 model, varying only the timing of the ESG-score change; all controls, fixed effects, and firm-clustered standard errors are as in Table IV. The lagged change spans t-2 to t-1, the movement observable when the letter is written. N = 447 firm-years (ESG share, 150 firms) and 436 (relative ESG, 149 firms).*

As Table XI shows, under the lagged specification, the coefficient on ESG Decline is small and insignificant for both the ESG share (0.0035,  $p = 0.352$ ) and relative ESG attention (0.0033,  $p = 0.154$ ). On the same reduced sample, the contemporaneous coefficient is also insignificant. Using the score change a CEO could actually observe therefore does not change the conclusion under either timing.

## 5.4 Separate Entry of the Two Change Measures

ESG Decline and ESG Improvement are negatively correlated ( $r = -0.27$ , Section 4.2) because a firm-year cannot simultaneously record a decline and an improvement. To confirm that entering them jointly does not distort the estimates, each H1 model is re-estimated with only one of the two measures. Table XII reports the coefficient of interest from each single-predictor model.

**Table XII.** H1 change measures entered separately.

| Model (dependent variable) | Variable             | coef (SE)        | p     | N   |
|----------------------------|----------------------|------------------|-------|-----|
| ESG share                  | ESG Decline only     | 0.0001 (0.0022)  | 0.965 | 596 |
| ESG share                  | ESG Improvement only | -0.0003 (0.0009) | 0.725 | 596 |
| log ESG share              | ESG Decline only     | 0.0002 (0.0012)  | 0.863 | 596 |
| log ESG share              | ESG Improvement only | -0.0003 (0.0006) | 0.606 | 596 |
| log relative ESG           | ESG Decline only     | 0.0008 (0.0007)  | 0.236 | 582 |
| log relative ESG           | ESG Improvement only | -0.0004 (0.0004) | 0.331 | 582 |

*Notes: Each row re-estimates an H1 model with only one of the two change measures and the full controls, fixed effects, and firm-clustered standard errors. Only the coefficient of interest is shown.*

Entering the measures separately leaves the conclusion unchanged. The coefficient on ESG Decline remains small and insignificant across all three dependent variables, and the same holds for ESG Improvement. The negative correlation between the two measures therefore

does not mask an effect that is concealed by their joint entry. The joint specification of the main models is retained.

## 5.5 Summary

Across every specification examined, no statistically significant association is detected between an ESG decline and ESG communication. This holds under three treatments of leverage, three functional forms of the dependent variable (raw, log, and cube-root), a subsample that removes the zero observations, separate entry of the two change measures, and an alternative timing of the score change. The conclusion also holds under random effects as well as fixed effects. The binary mention model, the timing-corrected model, and the quartile test in Section 4.6 point in the same direction. Taken together, the evidence consistently supports the conclusion that H1, H2a, and H2b are not supported.

## 6. Discussion and Conclusion

### 6.1 Discussion of Findings

This study asked how a decline in ESG performance reshapes the allocation of communicative attention across topics in CEO shareholder letters, and whether ownership structure moderates that response. Combining the attention-based view with legitimacy theory, it predicted that a publicly visible ESG decline would draw executive attention toward the threatened domain. CEOs were therefore expected to devote a greater share of the letter to ESG, both in absolute terms and relative to financial performance, diversity, and innovation (H1). This reallocation was expected to be stronger under high institutional ownership (H2a) and weaker under high firm-oriented ownership (H2b). None of these predictions is supported. No significant association is detected between an ESG decline and ESG attention, and neither ownership interaction is significant.

The null result for H1 is notable mainly for its consistency. The coefficient on ESG Decline is essentially zero for the log ESG share, for log relative ESG attention, and for the untransformed share reported as a robustness check. It also remains small and insignificant under a cube-root

transformation, in a subsample limited to firms that mention ESG, under three treatments of the leverage outlier, when the change measures are entered separately, and under random effects. The result is therefore not an artifact of one functional form, one influential observation, or one estimator. Rather than restating the results, the discussion turns to why a decline might leave the topical composition of the letter unmoved. Several mechanisms beyond those hypothesized are consistent with the evidence but were neither predicted nor tested.

A first possibility is that the CEO letter is simply not the channel through which a response to an ESG rating would surface. The letter is a curated, forward-looking document whose agenda is dominated by financial performance and strategy. A rating change may instead be addressed in the sustainability report, the proxy statement, or investor calls rather than in the one page the CEO devotes to framing the year. If so, the null reflects the choice of channel rather than the absence of a response. Topic attention in the letter would then not be a valid proxy for the executive attention considered by the theory. This reading has a direct methodological implication, developed in Section 6.2, and is the most important qualification of the study's central assumption.

A second possibility is timing. The rating for a fiscal year is finalized only after the letter is written, so the main models relate the letter to a verdict the CEO could not yet observe. The design addresses this by re-estimating every model with the lagged change a CEO could actually see. The coefficient on ESG Decline remains small and insignificant (Section 5.3), as does the effect on the decision to mention ESG (Table VI). The null therefore does not appear to be a timing artifact. Even so, the design links letters to score changes in general rather than to specific, dated downgrades. A sharper design that identified discrete downgrades by date and examined the letters written just afterwards might detect a response that this study cannot.

A third set of explanations concerns the trigger itself. A composite ESG rating is a gradual, aggregated, and somewhat opaque signal. Unlike a discrete controversy or scandal, a modest year-on-year change may simply not be salient enough to reallocate executive attention. The rating may therefore not function as the acute trigger assumed by the theory. It is also possible that ESG ratings were not yet salient enough to the investors these letters address over the 2018 to 2022 window for a rating movement to warrant a narrative response. Market and financial pressures on the letter's agenda may simply outweigh a rating signal. Finally, the direction of any relationship is genuinely ambiguous. Firms that say little about ESG may be those whose ratings subsequently fall, so reverse causality cannot be excluded. These readings are not

mutually exclusive, and the data do not decide among them. They matter because they locate the null in the nature of the trigger and the channel rather than in the study's execution. Measurement constraints may also contribute. The weak performance of the controls in the fixed-effects models, with firm size and the prior-year ESG score, both established predictors of ESG disclosure in cross-sectional work, becoming insignificant once firm heterogeneity is absorbed, suggests that the models capture little of the within-firm variation in topical attention. This leaves open the possibility that a response exists but operates at a level of granularity that keyword-based letter measurement does not detect. A further interpretation concerns the nature of the channel itself. The analysis has so far treated the CEO letter as a flexible, discretionary text whose composition can shift from year to year as circumstances change. The null invites a different reading. The letter may be a more institutionalized and less pliable genre than the impression-management literature typically assumes. CEO letters follow established conventions, frequently reuse the structure and language of prior years, and pass through investor-relations and legal review before publication. Their topical composition may therefore be shaped more by genre norms and continuity than by any single performance signal.

Institutional theory offers a useful perspective here. Meyer and Rowan (1977) describe how organizations adopt institutionalized forms ceremonially, so that visible structures become decoupled from underlying activity, while DiMaggio and Powell (1983) show how organizations converge on shared templates through isomorphic pressures rather than functional necessity. From this perspective, the stability of topic shares across ESG declines is not necessarily a failure of executives to attend to ESG. It may instead reflect the properties of a comparatively ritualized communicative form whose agenda is largely fixed in advance. The letter conforms to an established template, making its conformity more symbolic than a substantive response to the year's events. If the letter is partly ceremonial, its muted response to an ESG decline is less surprising. The discretion that makes the letter attractive to study may also be more constrained in practice than it appears in principle.

A final possibility is that firms respond to an ESG decline substantively rather than communicatively. A rating downgrade may prompt managerial action, including adjustments to operations, governance, remuneration, or supply-chain practices, that addresses the underlying weakness without a corresponding change in what the CEO writes. The response may therefore be real but behavioral rather than communicative. The firm may act without changing the content of the letter. This reverses the pattern often emphasized in the impression-

management literature, in which talk can substitute for action (Brennan & Merkl-Davies, 2013). Here, action may instead substitute for talk.

A related explanation concerns timing. An ESG development need not trigger an immediate change in the following letter. The response may still be taking shape, be reserved for a later reporting cycle, or executives may judge a single year's rating movement too transient to warrant a narrative reaction. The present design cannot distinguish between these mechanisms. Each, however, could produce the observed null without implying executive inattention to ESG. Future work should distinguish communicative silence from substantive response and examine whether responses are immediate or delayed.

The ownership hypotheses fare no better, and the reasons connect to the ownership literature on which the study draws. Neither interaction is significant. Although each sign lies in the predicted direction, an insignificant coefficient is not evidence that either type of owner conditions the relationship. Two features of the data reinforce this caution. Because H1 detects no baseline reallocation, there is in effect no main relationship for ownership to moderate. Firm-oriented ownership is also zero for roughly three-quarters of firm-years, so the H2b test draws on a minority of firms and has limited power. The broader implication for the ownership-and-disclosure literature is that the monitoring and governance differences that distinguish institutional from firm-oriented owners (Eng & Mak, 2003; Collin et al., 2017; Federo et al., 2020), while they shape formal disclosure and its tone, do not appear to extend to the discretionary topic composition of the CEO letter. This may be because there is no underlying attentional response for ownership to amplify or dampen. The ownership results are therefore best read as detecting no moderating association, not as showing that ownership is irrelevant to CEO communication in general.

One pattern runs counter to the hypotheses and constitutes a surprising finding. Among firms whose ESG score declined, those with the deepest declines write less about ESG. The pattern is concentrated on whether the topic is raised at all: firms in the most severe quartile of decline are about 31 percentage points less likely to mention ESG than those in the mildest quartile (Section 4.6), whereas the amount written once the topic is raised does not differ significantly. The direction is opposite to H1. This surprising finding runs counter to the theoretical expectation of Chapter 2. It is kept firmly exploratory for the three reasons given in Section 4.6. First, it is a contemporaneous association. Second, it does not survive the timing correction, under which it is instead a prior improvement that predicts silence. Third, its direction cannot



be established. The finding also stands in tension with the disclosure research that motivated the study, in which weaker performers tend to disclose more, not less (Patten, 2002; Cong et al., 2014). A likely reason is scope. That work examines environmental disclosure as a single domain, whereas this study measures the share of a bounded letter given to a composite ESG construct that competes with three other topics for the same limited space.

## 6.2 Theoretical and Practical Implications

The findings refine rather than confirm the study's central idea. Extending the attention-based view to what CEOs communicate implies that a salient, externally imposed signal should surface in the topical composition of the letter. Here it does not. Topic-level ESG attention is unrelated to changes in the ESG score. Two readings are consistent with the evidence, and the data do not decide between them. An ESG rating change may not be the acute trigger assumed by the theory for these firms, but rather a gradual and diffuse signal. Alternatively, a topic-level response may exist but remain invisible in the share of a bounded letter, because attention across these topics is not strictly zero-sum, as the correlation evidence only partly supports. Either way, the assumption that discretionary communication mirrors shifts in signalled performance needs qualification.

The most consequential theoretical implication concerns the status of the CEO letter as a measure of executive attention. Much impression-management and attention research treats the letter as a window onto what occupies the executive mind (Hooghiemstra, 2000; Cho & Hambrick, 2006). The null here, together with the possibility that any ESG response is voiced through other channels, suggests that the letter may be a valid proxy for executive attention on some dimensions but not for externally imposed, non-financial performance signals. The natural home of such responses may instead be the sustainability report or the proxy statement. For the impression-management literature specifically, there is no evidence that a decline triggers active thematic engagement with ESG. If anything, the exploratory pattern points cautiously toward omission rather than the thematic manipulation documented in this literature (Brennan & Merkl-Davies, 2013). Future theoretical work could therefore treat the choice of communicative channel as a strategic variable rather than assuming that the letter carries every response.

These results also speak to legitimacy theory as an account of communicative response. Legitimacy theory holds that a firm facing a legitimacy gap acts to repair its public standing

(Suchman, 1995), which would be expected to surface in what the firm chooses to communicate. The evidence, however, does not support this expectation for the CEO letter. A publicly visible ESG decline does not move the topical composition of the letter. Legitimacy repair, if it occurs, may therefore operate through channels other than discretionary executive communication, such as audited sustainability reporting. Read in this way, the null does more than leave the hypotheses unconfirmed. It helps specify the conditions under which each theory operates. For the attention-based view, the results suggest that an external stimulus may reliably capture and redirect finite executive attention only when it has particular properties. A discrete, self-generated, and unambiguous event, such as a controversy, a lawsuit, or a major acquisition, fits the theory's original logic of attention being drawn to salient stimuli (Ocasio, 1997; Cho & Hambrick, 2006). A composite ESG rating has almost the opposite properties. It is gradual, aggregated, externally assigned, and revised in small increments. The boundary condition identified here therefore concerns stimulus salience and provenance. The attention-based view predicts reallocation in response to acute and internally salient signals, but not necessarily to diffuse, third-party performance scores, even when those scores are publicly visible. This qualification is consistent with the communicative development of the theory, which stresses that attention is enacted through particular vocabularies and channels rather than responding automatically to any external cue (Ocasio, Laamanen & Vaara, 2018).

For legitimacy theory, the boundary condition concerns the channel through which repair is communicated. The theory predicts that a firm facing a legitimacy gap will act to repair its standing (Suchman, 1995), and prior work shows such repair surfacing in disclosure (Bansal & Clelland, 2004; Patten, 2002). The present evidence does not so much contradict this logic as help locate it. If legitimacy repair is channel-specific, audited sustainability reports and proxy statements may carry the response for ESG, while the CEO letter may be used mainly for the financial framing of the year. The absence of a response in the letter would then not be evidence against legitimacy theory. It would instead indicate where repair is communicated. The theory may still apply, but more selectively across the channels that the relevant audience monitors for a given domain. Framing the null in terms of boundary conditions rather than disconfirmation is therefore an important theoretical contribution of the study. It identifies limits to the application of two widely used theories rather than simply showing that their predictions did not appear in one setting.

The practical implications follow directly and counsel caution. Stakeholders sometimes read the prominence a letter gives to ESG as a sign of how seriously a firm takes its ESG standing. The evidence gives no basis for this inference, since letter-level ESG attention does not move with the ESG score. The exploratory pattern also suggests that the presence or absence of an ESG mention may even mislead. Firms whose scores fell most sharply were, if anything, the quieter ones. Investors and analysts are therefore better served by the ratings themselves and by audited disclosures than by the discretionary emphasis of the letter. Methodologically, the study also shows that impression management can be studied through the distribution of topic attention, not sentiment alone. A well-identified null against a strong theoretical prior is itself an informative result.

### **6.3 Limitations and Future Research**

Several limitations bound these conclusions and point to possible extensions. The measure captures how much a topic is raised, but not how it is framed, how sincerely, or with what valence. A firm could raise ESG defensively without the measure distinguishing this from genuine engagement. The dependent variable is also a composite ESG construct. The environmental, social, and governance pillars may trigger distinct responses that aggregation hides. There is evidence that the three pillars are priced differently (Wijeweera, Kortt & Kim, 2026), which makes a pillar-level analysis a natural next step. Future work could decompose the trigger to the pillar level and combine topic attention with sentiment to capture both what is said and how it is said. Keyword-based measurement could also be complemented with embedding- or language-model-based approaches. Another step would be to compare gradual score changes with discrete, salient events such as controversies or scandals. These may act as the acute trigger that a composite rating does not. Research could also examine channels beyond the letter, such as earnings calls or sustainability reports, where a response not carried by the letter may instead appear.

The sample also bounds the reach of the findings. It comprises large, established, US-listed firms over a five-year window in which declines were the minority experience and firm-oriented ownership was concentrated in a small share of firm-years. This limits the power of the H2b test. The results may therefore not extend to smaller firms, other periods, or other institutional settings. A particularly promising direction is to repeat the analysis where the cultural and institutional context differs markedly from the United States, such as China, Japan,

or India. Legitimacy pressures and the norms of executive communication are culturally shaped, so the way firms narrate an ESG decline may differ across settings. US and Japanese CEOs, for instance, already differ in how they explain good and bad news in their letters, particularly after performance has deteriorated (Hooghiemstra, 2010). Such work would show whether the null reflects something specific to the US context or holds more broadly. It would also test the generalisability of both the null and the exploratory pattern.

## 6.4 Conclusion

This study offers a first systematic account of whether CEOs redirect the attention of their letters toward ESG when their ESG performance declines. The results indicate that they do not. No statistically significant association emerges across the specifications and robustness checks, and the null also holds when the timing of the score change is corrected and when the estimator is changed. The contribution is twofold. The study provides a well-identified null result against a strong theoretical prior and introduces a method for studying impression management through the distribution of topic attention rather than sentiment alone. This relational, allocation-based view of topic attention is the study's distinctive contribution. It treats the CEO letter as a bounded space in which ESG, financial performance, innovation, and diversity compete for attention. This perspective remains useful whether future studies find an effect or, as in the present study, no significant relationship. The null is itself informative because it places limits on where the attention-based view applies and raises questions about whether the CEO letter is a valid channel for externally imposed non-financial signals. One post-hoc pattern departs from the null: firms with the deepest declines are the least likely to mention ESG at all. This finding remains exploratory and is offered as a direction for future research rather than as a conclusion.

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## Appendix

### Appendix A. Keyword Lists

The four lists below are the result of the construction and validation procedure described in Section 3.2. For each topic, the terms are reported in two groups. The first group contains terms that occurred in the 20 pilot letters and met the specificity criterion. Following the general principle of evaluating candidate words in their sample context (Loughran & McDonald, 2011), a term was retained only where its usage clearly signalled the intended topic, and was excluded where it was predominantly ambiguous or belonged to a different construct. The second group contains further candidate terms that did not occur in the pilot letters but were retained on content grounds, since absence from 20 letters is not evidence that a term is unsuitable for a corpus of 750. Terms are listed alphabetically within each group and matching is case-insensitive.

| Topic                 | Observed | Not observed | Total |
|-----------------------|----------|--------------|-------|
| ESG                   | 31       | 44           | 75    |
| Diversity             | 24       | 35           | 59    |
| Innovation            | 27       | 14           | 41    |
| Financial performance | 25       | 10           | 35    |
| Total                 | 107      | 103          | 210   |

#### A.1 ESG

Source lists: environment and governance subcategories of Ahn (2022) and the ESG word list of Baier, Berninger, and Kiesel (2020), supplemented inductively.

##### Terms observed in the pilot sample (31)

Atmosphere, carbon, carbon footprint, carbon-neutral, CDP, clean energy, cleaner, climate, climate change, deforestation, emissions, energy efficiency, environmental, epa, ESG, fossil, governance, green, greenhouse, GRESB, GRI, recycled, recycling, renewable, Science Based Targets, solar, sustainability, sustainable, sustainably, waste, water consumption,

##### Terms from published sources not observed in the pilot sample (44)

anti-trust, biodegradable, biodiversity, biofuel, biofuels, clean-up, cleanup, codes of conduct, contaminated, contamination, corruption, decarbonization, ecofriendly, ecological, emission, emit,

ethics, freshwater, GHG, GHGs, groundwater, misconduct, net zero, nitrogen, non-executive, ozone, pesticide, pesticides, pollutants, pollution, recycle, Scope 1, Scope 2, Scope 3, shareholder resolutions, species, superfund, warming, wastes, water usage, wetlands, wilderness, wildlife, work councils

## **A.2 Diversity**

Source lists: diversity subcategory of Baier, Berninger, and Kiesel (2020) and the DEI word list of Demers, Wang, and Wu (2024), supplemented inductively.

### **Terms observed in the pilot sample (24)**

belonging, black, Chief Diversity Officer, corporate equality index, DE&I, diversity, Employee Resource Groups, equality, equality index, ergs, gender, gender parity, inclusion, lgbtq+, minorities, minority, racial, racial equity, racial justice, social justice, unconscious bias, unconscious bias training, underrepresented, women

### **Terms from published sources not observed in the pilot sample (35)**

allyship, bisexual, DEI, disability equality index, equal opportunity, equal opportunity employer, ethnically, ethnicities, ethnicity, female, females, gay, gays, gender pay, genders, historically black college, homosexual, lesbian, lesbians, lgbt, national origin, neurodiversity, non-discrimination, pay equity, pay gap, people of color, protected class, religion, religious, sexual orientation, systemic racism, transgender, veteran-owned, woman, workforce diversity

## **A.3 Innovation**

Source list: research and development glossary of Merkley (2014), supplemented inductively.

### **Terms observed in the pilot sample (27)**

5G, AI, APIs, artificial intelligence, automation, breakthrough, cloud, cloud-based, cybersecurity, data center, digital infrastructure, digital transformation, e-commerce, ecommerce, fintech, innovate, innovation, innovations, innovative, machine learning, new product, product development, robotic, robotics, robots, software, technologies

### **Terms from published sources not observed in the pilot sample (14)**

algorithm, API, emerging technology, patent, patents, process amelioration, product engineering, production processes, put into production, R&D, research and development, semiconductor, technology breakthrough, technology investment

## **A.4 Financial Performance**

Source lists: financial-metric terms from the FinTextSim keyword list of Jehnen, Ordieres-Meré and Villalba-Díez (2025), which builds on the topic scheme of Li (2010) and the word2vec-based list of Fengler and Phan (2025); remaining terms developed inductively from the pilot sample.

**Terms observed in the pilot sample (25)**

basis points, book value, cash flow, cash flows, dividend, dividends, earnings, earnings per share, EBITDA, EPS, financial performance, financial results, free cash flow, gross profit, margin, margins, net income, per share, profit, profitability, profits, revenue growth, ROA, shareholder return, shareholder returns

**Candidate terms not observed in the pilot sample (10)**

annual results, EBDA, EBIT, net revenue, operating margin, profit margin, quarterly results, return on assets, return on equity, ROE

**A.5 Matching Rules**

Terms are matched as whole words. Where a term is valid but carries a documented false-positive context in the pilot corpus, that context is excluded from the search pattern rather than the term being discarded. Four such exclusions apply: recycling does not match recycling capital, profits does not match not-for-profits, minority does not match minority interest, and black does not match in the black.

Within each topic, all terms are combined into a single search pattern in which longer phrases take precedence over shorter ones, so that each occurrence in the text is counted only once even where it would satisfy several list entries. Examples include earnings per share and earnings, gross profit and profit, free cash flow and cash flow, carbon-neutral and carbon, racial justice and racial, and product development and new product; the rule applies to every such overlap within a topic, not only these.

**Appendix B. Sample Composition and Missing CEO Letters****Sample Composition**

The sample comprises the 150 firms selected as described in Section 3.1, observed over 2018 to 2022, which yields 750 potential firm-year observations; excluding the 2018 base year, on which the year-on-year change measure cannot be computed, leaves an estimation sample of 600 firm-years. A CEO letter could not be obtained for ten firm-years, listed below, of which six fall in the base year and four within the estimation window, so that the four topic shares are available for 596 firm-years.

**Missing CEO Letters**

| ISIN         | Year | Firm                    |
|--------------|------|-------------------------|
| US4432011082 | 2018 | Howmet Aerospace Inc    |
| US5404241086 | 2018 | Loews Corporation       |
| US62482R1077 | 2018 | Mr Cooper Group Inc     |
| US89832Q1094 | 2018 | Truist Financial Corp   |
| US1976411033 | 2018 | Columbia Financial Inc  |
| US8115431079 | 2018 | Seaboard Corp           |
| US62482R1077 | 2019 | Mr Cooper Group Inc     |
| US00507V1098 | 2022 | Activision Blizzard Inc |
| US6936561009 | 2022 | PVH Corporation         |
| US1488061029 | 2022 | Catalent Inc            |

**Appendix C. Variable Definitions**

| Variable                          | Definition                                                                                        | Source      |
|-----------------------------------|---------------------------------------------------------------------------------------------------|-------------|
| ESG topic share                   | Share of ESG terms in total letter words $\times 100$                                             | CEO letters |
| Diversity topic share             | Share of diversity terms in total letter words $\times 100$                                       | CEO letters |
| Financial performance topic share | Share of financial performance terms in total letter words $\times 100$                           | CEO letters |
| Innovation topic share            | Share of innovation terms in total letter words $\times 100$                                      | CEO letters |
| Relative ESG attention            | ESG topic share divided by the sum of the four topic shares                                       | CEO letters |
| ESG Decline                       | Absolute value of the year-on-year percentage change in the ESG score if negative, zero otherwise | TRESGS      |
| ESG Improvement                   | Year-on-year percentage change in the ESG score if positive, zero otherwise                       | TRESGS      |
| Institutional ownership           | Percentage of shares held by investment companies                                                 | NOSHIC      |

|                             |                                                                   |                 |
|-----------------------------|-------------------------------------------------------------------|-----------------|
| Firm-oriented ownership     | Percentage of strategic holdings not held by investment companies | NOSHST – NOSHIC |
| Firm size                   | Natural logarithm of total assets                                 | WC02999         |
| Prior financial performance | Return on assets in the year prior to the CEO letter              | WC08326         |
| Prior ESG score level       | ESG score in the previous year                                    | TRESGS          |
| Financial leverage          | Total debt to common equity                                       | WC08231         |
| Year fixed effects          | Indicator variables for each observation year                     | Constructed     |

## Appendix D. Declaration on the Use of Artificial Intelligence

While writing this thesis, I used AI-based tools as supportive aids in several phases of the writing and revision process. Their use was limited to the following purposes: improving the clarity, conciseness, and academic style of the text I wrote myself; validating the keyword lists for content analysis (see explanation in Section 3.2); assisting in checking my R Studio code for completeness; and a final review of the document for formatting errors. The entire research design, data collection, statistical analysis, interpretation of the results, and final formulation were carried out and critically reviewed by me personally. AI tools were not used to independently perform the statistical analysis or to generate content.

Examples of prompts used:

- “Check the clarity and academic style of this paragraph without changing the content, argumentation, or citations.”
- “Search the 20 pilot letters using regular expressions for all occurrences of the term [X] and show me the respective context so that I can assess whether it is a genuine mention of the topic or a false positive.”
- “Go through this R code and check whether it includes all the necessary steps for the regression analysis.”
- “Search the entire document for formatting errors like inconsistent citation styles, incorrect chapter numbering, missing cross-references.”